2026 Fall Linear Algebraic Groups Note

Ridongen ridongen031110@gmail.com

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Introduction

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2. [Wat79] W. C. Waterhouse, *Introduction to Affine Group Schemes*, GTM 66.
3. [Dem70] *SGA III: Schémas en groupes*.
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Goal. Understand the structure of linear algebraic groups (LAG) over a field k.

Remark. Faithfully follow Miaofen Chen's lecture in Fudan University, 2026 Fall.

1 Definition and basic properties

1.1 Definitions and conventions

Let k be a field. For a scheme X of finite type over k, let

$$X^{\mathrm{cl}} := \{\text{closed points of } X\}, \qquad j : X^{\mathrm{cl}} \hookrightarrow X, \qquad \mathcal{O}_{X^{\mathrm{cl}}} := j^{-1}\mathcal{O}_X.$$

The locally ringed spaces obtained in this way are the *ultra-schemes* of [Gro66, §10.9] For an affine scheme,

$$(\operatorname{Spec} A)^{\mathrm{cl}} = \operatorname{Spm} A,$$

the maximal spectrum of A.

Proposition 1.1. [Gro66, 10.9.6] The functor $X \mapsto X^{\mathrm{cl}}$, with the induced structure sheaf, gives an equivalence between the category of Jacobson schemes and the category of ultra-schemes.

Convention. From now on, schemes over k are schemes of finite type over k. Under the preceding equivalence, affine schemes are written as $\operatorname{Spm} A$. Thus a scheme over k admits a finite affine covering

$$X = \bigcup_{i=1}^{r} \operatorname{Spm} A_i.$$

Definition 1.2. An *algebraic group over k* is a scheme G/k together with morphisms

$$m : G \times_k G \longrightarrow G, \qquad e : \operatorname{Spec} k \longrightarrow G, \qquad i : G \longrightarrow G,$$

called multiplication, unit, and inverse, satisfying the following properties.

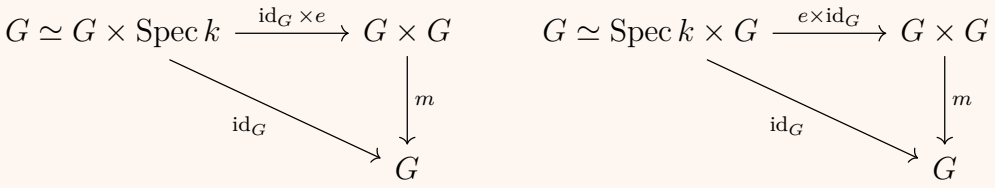
(1) Associativity.

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{m \times \mathrm{id}_G} & G \times G \\ {\scriptstyle \mathrm{id}_G \times m}\downarrow & & \downarrow{\scriptstyle m} \\ G \times G & \xrightarrow[m]{} & G \end{array}$$

(2) Existence of a unit. Writing $p_G : G \to \operatorname{Spec} k$ for the structure morphism,

$$m \circ (\mathrm{id}_G, e \circ p_G) = \mathrm{id}_G = m \circ (e \circ p_G, \mathrm{id}_G).$$

Equivalently, the two diagrams

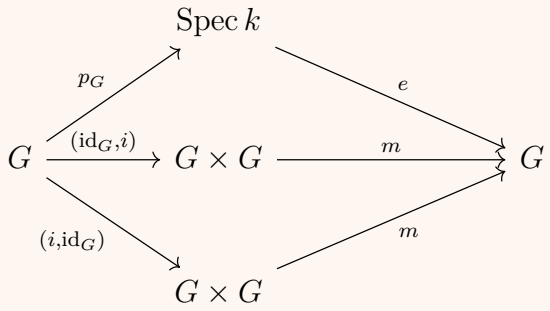


commute.

(3) Inverse property.

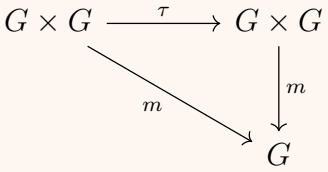
$$m \circ (\mathrm{id}_G, i) = e \circ p_G = m \circ (i, \mathrm{id}_G).$$

Equivalently,



Definition 1.3. A *linear algebraic group over* k is an algebraic group over k whose underlying scheme is affine.

Definition 1.4. An algebraic group G is *commutative* if the diagram



$$\big(\tau(x, y) = (y, x)\big)$$

commutes.

Remark. Let $G = \operatorname{Spm} A$, where A is a finitely generated commutative k-algebra. The structure morphisms of G correspond contravariantly to

$$m^{\#} : A \longrightarrow A \otimes_k A, \qquad e^{\#} : A \longrightarrow k, \qquad i^{\#} : A \longrightarrow A.$$

These are called, respectively, the *comultiplication*, *counit*, and *coinverse* (or *antipode*).

Moreover,

$$A \simeq \mathrm{Hom}_{k\text{-schemes}}(G, \mathbb{A}^1_k).$$

If

$$m^{\#}(f) = \sum_j f_j \otimes g_j,$$

then for $x, y \in G$,

$$f\big(m(x,y)\big) = \sum_j f_j(x) g_j(y).$$

The group axioms translate into the following identities for $m^{\#}, e^{\#}, i^{\#}$.

Coassociativity.

$$(m^{\#} \otimes \mathrm{id}_A) \circ m^{\#} = (\mathrm{id}_A \otimes m^{\#}) \circ m^{\#}.$$

Equivalently,

$$\begin{array}{ccc} A \otimes A \otimes A & \xleftarrow{\mathrm{id}_A \otimes m^{\#}} & A \otimes A \\ \uparrow{\scriptstyle m^{\#} \otimes \mathrm{id}_A} & & \uparrow{\scriptstyle m^{\#}} \\ A \otimes A & \xleftarrow[m^{\#}]{} & A \end{array}$$

Counit. Using $k \otimes A \simeq A \simeq A \otimes k$,

$$(e^{\#} \otimes \mathrm{id}_A) \circ m^{\#} = \mathrm{id}_A = (\mathrm{id}_A \otimes e^{\#}) \circ m^{\#}.$$

Equivalently,

$$\begin{array}{ccc} A & \xleftarrow{\mathrm{id}_A \otimes e^{\#}} & A \otimes A \\ \uparrow{\scriptstyle e^{\#} \otimes \mathrm{id}_A} & & \uparrow{\scriptstyle m^{\#}} \\ A \otimes A & \xleftarrow[m^{\#}]{} & A \end{array}$$

Antipode identity. Let

$$\mu_A : A \otimes_k A \longrightarrow A$$

be multiplication in A, and let

$$\eta_A : k \longrightarrow A$$

be the unit map. Then

$$\mu_A \circ (i^{\#} \otimes \mathrm{id}_A) \circ m^{\#} = \eta_A \circ e^{\#} = \mu_A \circ (\mathrm{id}_A \otimes i^{\#}) \circ m^{\#}.$$

Definition 1.5. A commutative k-algebra A, together with homomorphisms

$$m^{\#} : A \to A \otimes_k A, \qquad e^{\#} : A \to k, \qquad i^{\#} : A \to A$$

satisfying the preceding coassociativity, counit, and antipode identities, is called a *Hopf algebra*.

Remark (Anti-equivalence of categories)**.** There is an anti-equivalence

$$\left\{ \begin{array}{c} \text{finitely generated commutative} \\ \text{Hopf } k\text{-algebras} \end{array} \right\} \simeq \{\text{linear algebraic groups over } k\},$$

sending A to $\operatorname{Spm} A$.

Definition 1.6. A *homomorphism of algebraic groups* $\varphi : G \to G'$ is a morphism of schemes such that

$$\begin{array}{ccc} G \times G & \xrightarrow{(\varphi,\varphi)} & G' \times G' \\ \downarrow{\scriptstyle m_G} & & \downarrow{\scriptstyle m_{G'}} \\ G & \xrightarrow[\varphi]{} & G' \end{array}$$

commutes.

Definition 1.7. An *algebraic subgroup* H/k of an algebraic group G/k is an algebraic group H/k such that

1. H is a closed k-subscheme of G;
2. the inclusion $H \hookrightarrow G$ is a homomorphism of algebraic groups.

Remark. In the preceding definition, the word "closed" in (1) is redundant for algebraic subgroup schemes of finite type over a field.

1.1.1 Algebraic groups as functors

Let Alg_k denote the category of finitely generated commutative k-algebras.

Proposition 1.8 (Yoneda)**.** The functor-of-points construction

$$X \longmapsto \big(R \longmapsto X(R) := \mathrm{Hom}_{\mathrm{Sch}\,/k}(\operatorname{Spec} R, X)\big)$$

gives a fully faithful functor

$$\{\text{schemes over } k\} \longrightarrow \{\text{functors } \mathrm{Alg}_k \to \mathbf{Set}\}.$$

The functors in its essential image are called *representable functors*.

Now let G be an algebraic group over k, with multiplication m, unit e, and inverse i. For every $R \in \mathrm{Alg}_k$, these induce

$$m_R : G(R) \times G(R) \longrightarrow G(R), \qquad e_R := e \circ p_R \in G(R), \qquad i_R : G(R) \longrightarrow G(R),$$

where $p_R : \operatorname{Spec} R \to \operatorname{Spec} k$ is the structure morphism. The commutative diagrams in the definition of an algebraic group imply that $(G(R), m_R)$ is a group.

Hence an algebraic group gives a functor

$$\widetilde{G} : \mathrm{Alg}_k \longrightarrow \mathbf{Grp}, \qquad R \longmapsto G(R).$$

Proposition 1.9 (Functorial definition)**.** An algebraic group over k can equivalently be described as a functor

$$\widetilde{G} : \mathrm{Alg}_k \longrightarrow \mathbf{Grp}$$

such that the composite

$$\mathrm{Alg}_k \xrightarrow{\widetilde{G}} \mathbf{Grp} \longrightarrow \mathbf{Set}$$

is representable.

Exercise 1.10. Check directly that the scheme-theoretic and functorial definitions of an algebraic group coincide.

Remark. For an algebraic group, the unit e and inverse i are uniquely determined by the multiplication m.

Remark. An algebraic group G is commutative if and only if $G(R)$ is a commutative group for every $R \in \mathrm{Alg}_k$.

If k is perfect and G is reduced (equivalently, by the smoothness criterion below, G is smooth), then it is enough to test this on a separable closure k^s:

$$G \text{ is commutative} \iff G(k^s) \text{ is commutative.}$$

Example 1.11 (Basic examples)**.** 1. The trivial group:

$$G = \operatorname{Spm} k.$$

2. The additive group:

$$\mathbb{G}_a : \mathrm{Alg}_k \longrightarrow \mathbf{Grp}, \qquad R \longmapsto (R, +),$$

with

$$\mathbb{G}_a = \operatorname{Spm} k[T].$$

Its Hopf algebra structure is

$$m^{\#} : k[T] \longrightarrow k[T] \otimes_k k[T], \qquad T \longmapsto T \otimes 1 + 1 \otimes T,$$

$$e^{\#} : k[T] \longrightarrow k, \qquad T \longmapsto 0,$$

and

$$i^{\#} : k[T] \longrightarrow k[T], \qquad T \longmapsto -T.$$

Thus, if $m^{\#}(T) = \sum_j f_j \otimes g_j$, then for $x, y \in \mathbb{G}_a$,

$$\sum_j f_j(x) g_j(y) = x + y.$$

3. The multiplicative group:

$$\mathbb{G}_m : \mathrm{Alg}_k \longrightarrow \mathbf{Grp}, \qquad R \longmapsto (R^{\times}, \cdot),$$

with

$$\mathbb{G}_m = \operatorname{Spm} k[T, T^{-1}].$$

Its Hopf algebra structure is

$$m^{\#} : k[T, T^{-1}] \longrightarrow k[T, T^{-1}] \otimes_k k[T, T^{-1}], \qquad T \longmapsto T \otimes T,$$

$$e^{\#} : k[T, T^{-1}] \longrightarrow k, \qquad T \longmapsto 1,$$

and

$$i^{\#} : k[T, T^{-1}] \longrightarrow k[T, T^{-1}], \qquad T \longmapsto T^{-1}.$$

4. Let Γ be a finite group. The associated constant algebraic group is

$$\Gamma = \coprod_{\gamma\in\Gamma} \operatorname{Spec} k = \operatorname{Spm}\left(\bigoplus_{\gamma\in\Gamma} k e_\gamma\right),$$

where the e_γ are orthogonal idempotents. Writing $\mathcal{O}(\Gamma) = \bigoplus_{\gamma\in\Gamma} k e_\gamma$, one has

$$m^{\#} : \mathcal{O}(\Gamma) \longrightarrow \mathcal{O}(\Gamma) \otimes_k \mathcal{O}(\Gamma), \qquad e_x \longmapsto \sum_{g\in\Gamma} e_g \otimes e_{g^{-1}x},$$

$$e^{\#} : \mathcal{O}(\Gamma) \longrightarrow k, \qquad e_x \longmapsto \begin{cases} 1, & x = e, \\ 0, & x \neq e, \end{cases}$$

and

$$i^{\#} : \mathcal{O}(\Gamma) \longrightarrow \mathcal{O}(\Gamma), \qquad e_x \longmapsto e_{x^{-1}}.$$

5. For $n \in \mathbb{N}$,

$$\mathrm{GL}_n \supseteq B_n \supseteq U_n, \qquad T_n \subseteq B_n.$$

As functors on Alg_k,

$$\mathrm{GL}_n(R) = \left\{(x_{ij}) \in M_n(R) \,\middle|\, \det(x_{ij}) \in R^\times\right\},$$

$$B_n(R) = \left\{(x_{ij}) \in \mathrm{GL}_n(R) \,\middle|\, x_{ij} = 0 \text{ for } i > j\right\},$$

$$U_n(R) = \left\{(x_{ij}) \in B_n(R) \,\middle|\, x_{ii} = 1 \text{ for all } i\right\},$$

and

$$T_n(R) = \left\{(x_{ij}) \in B_n(R) \,\middle|\, x_{ij} = 0 \text{ for } i < j\right\}.$$

Thus the typical matrix shapes are

$$B_n : \begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix}, \qquad U_n : \begin{pmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{pmatrix}, \qquad T_n : \begin{pmatrix} * & 0 & 0 \\ 0 & * & 0 \\ 0 & 0 & * \end{pmatrix}.$$

Here GL_n is a reductive group, B_n is a solvable (Borel, hence parabolic) subgroup of GL_n, U_n is a unipotent group, and T_n is a torus.

1.2 Basic properties

Proposition 1.12. Algebraic groups are separated.

Proof. It is enough to show that the diagonal

$$\Delta_G = \{(x, x) \mid x \in G\} \subseteq G \times G$$

is closed. Consider the morphism

$$G \times G \xrightarrow{(\mathrm{id}_G, i)} G \times G \xrightarrow{m} G, \qquad (x, y) \longmapsto xy^{-1}.$$

Then

$$\Delta_G = \big(m \circ (\mathrm{id}_G, i)\big)^{-1}(e),$$

and the identity section $e : \operatorname{Spec} k \to G$ is a closed point. Hence Δ_G is closed. □

Proposition 1.13. Let G be an algebraic group over k. Then

$$G \text{ is geometrically reduced} \quad \Longleftrightarrow \quad G \text{ is smooth over } k.$$

Proof. The implication "smooth ⇒ geometrically reduced" is immediate.

For the converse, after extending scalars to an algebraic closure $\overline{k}$, we may assume $k = \overline{k}$. Then G is reduced. The smooth locus of the finite-type k-scheme G is a nonempty open dense subset. Choose a smooth point $g \in G$. For any $h \in G$, left translation by hg^{-1} is an automorphism of G sending g to h. Smoothness is preserved by isomorphisms, so every point of G is smooth. Thus G is smooth. □

Remark. If $\operatorname{char} k = 0$, then every algebraic group over k is smooth.[1]

Example 1.14. Assume $\operatorname{char} k = p > 0$. Define

$$\mu_p := \ker\Big(\mathbb{G}_m \xrightarrow{[p]} \mathbb{G}_m\Big), \qquad x \longmapsto x^p.$$

[1] 2026.9.7 stopped at here.

Equivalently, μ_p is the fiber product

$$\begin{array}{ccc} \mu_p & \longrightarrow & \operatorname{Spec} k \\ \downarrow & & \downarrow{\scriptstyle e} \\ \mathbb{G}_m & \xrightarrow{[p]} & \mathbb{G}_m. \end{array}$$

On coordinate rings, the p-power map corresponds to

$$k[T, T^{-1}] \longrightarrow k[T, T^{-1}], \qquad T \longmapsto T^p,$$

while the identity section corresponds to

$$k[T, T^{-1}] \longrightarrow k, \qquad T \longmapsto 1.$$

Hence

$$\mu_p \simeq \operatorname{Spm} \frac{k[T]}{(T^p - 1)} = \operatorname{Spm} \frac{k[T]}{((T-1)^p)}.$$

Thus μ_p is nonreduced, and therefore it is not smooth.

Remark (Reduced subscheme)**.** For a scheme X, write X_{red} for its reduced closed subscheme. If $X = \operatorname{Spec} A$, then

$$X_{\mathrm{red}} = \operatorname{Spec}\big(A/\sqrt{(0)}\big) \hookrightarrow \operatorname{Spec} A.$$

The universal property used below is that every morphism from a reduced scheme to X factors uniquely through X_{red}.

Recall. A morphism $\varphi : G \to H$ of algebraic groups is a homomorphism precisely when the multiplication square commutes:

$$\begin{array}{ccc} G \times G & \xrightarrow{m_G} & G \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ H \times H & \xrightarrow[m_H]{} & H. \end{array}$$

Idea. Restrict the multiplication of G to

$$G_{\mathrm{red}} \times G_{\mathrm{red}} \longrightarrow G.$$

To factor this morphism through G_{red}, it is enough that $G_{\mathrm{red}} \times G_{\mathrm{red}}$ be reduced. In particular, this holds when G_{red} is geometrically reduced.

Proposition 1.15. Let G be an algebraic group over k. If G_{red} is geometrically reduced, then G_{red} is an algebraic subgroup of G. In particular, this holds whenever k is perfect.

Proof. Let $j : G_{\mathrm{red}} \hookrightarrow G$ be the canonical closed immersion.

Since $\operatorname{Spec} k$ is reduced, the identity morphism $e : \operatorname{Spec} k \to G$ factors uniquely through G_{red}:

$$\begin{array}{ccc} \operatorname{Spec} k & \xrightarrow{\quad e \quad} & G \\ & {\scriptstyle e_{\mathrm{red}}} \searrow \quad \nearrow {\scriptstyle j} & \\ & G_{\mathrm{red}} & \end{array}$$

Since G_{red} is reduced, the inverse morphism $\iota \circ j : G_{\mathrm{red}} \to G$ factors uniquely through G_{red}.

Finally, because G_{red} is geometrically reduced, $G_{\mathrm{red}} \times_k G_{\mathrm{red}}$ is reduced. Hence

$$m_G \circ (j \times j) : G_{\mathrm{red}} \times_k G_{\mathrm{red}} \longrightarrow G$$

factors uniquely through G_{red}:

$$\begin{array}{ccc} G_{\mathrm{red}} \times G_{\mathrm{red}} & \overset{m_{\mathrm{red}}}{\dashrightarrow} & G_{\mathrm{red}} \\ {\scriptstyle j\times j}\downarrow & & \downarrow{\scriptstyle j} \\ G \times G & \xrightarrow[m_G]{} & G. \end{array}$$

Thus the identity, inverse, and multiplication of G restrict to G_{red}, so G_{red} is an algebraic subgroup of G.

If k is perfect, every reduced finite-type k-scheme is geometrically reduced by [GW20, Proposition 5.49], so the hypothesis is automatic. □

Example 1.16 (Why geometric reducedness matters)**.** Let

$$k = \mathbb{F}_p(t), \qquad K = k[x]/(x^p - t), \qquad X = \operatorname{Spec} K.$$

Since $t \notin k^p$, the polynomial $x^p - t$ is irreducible over k, so K is a field and X is reduced. If $\beta \in \overline{k}$ satisfies $\beta^p = t$, then

$$X_{\overline{k}} \cong \operatorname{Spec} \frac{\overline{k}[y]}{(y^p - t)} = \operatorname{Spec} \frac{\overline{k}[y]}{(y - \beta)^p},$$

which is not reduced. Thus X is reduced but not geometrically reduced.

If $\alpha \in K$ is the class of x, so that $\alpha^p = t$, then

$$X \times_k X \cong \operatorname{Spec}(K \otimes_k K) \cong \operatorname{Spec} \frac{K[y]}{(y^p - t)} = \operatorname{Spec} \frac{K[y]}{(y - \alpha)^p},$$

which is not reduced.

Definition 1.17 (Kernel)**.** Let $\varphi : G \to H$ be a homomorphism of algebraic groups. Its kernel is the fiber product

$$\ker(\varphi) = G \times_{H,e} \operatorname{Spec} k,$$

i.e.

$$\begin{array}{ccc} \ker(\varphi) & \longrightarrow & \operatorname{Spec} k \\ \downarrow & & \downarrow e \\ G & \xrightarrow[\varphi]{} & H. \end{array}$$

Definition 1.18 (Normal algebraic subgroup)**.** Let G be an algebraic group over k, and let $H \subseteq G$ be an algebraic subgroup. Then H is *normal* if, for every $R \in \mathrm{Alg}_k$,

$$H(R) \trianglelefteq G(R).$$

Remark. Let $\varphi : G \to H$ be a homomorphism. For every $R \in \mathrm{Alg}_k$,

$$(\ker \varphi)(R) = \ker\big(G(R) \longrightarrow H(R)\big).$$

Hence $\ker \varphi$ is a normal algebraic subgroup of G.

Remark. If G and H are affine, then $\ker \varphi$ is affine. Writing

$$I_H = \ker\big(e^\# : \mathcal{O}(H) \to k\big), \qquad \varphi^\# : \mathcal{O}(H) \to \mathcal{O}(G),$$

one has

$$\mathcal{O}(\ker \varphi) \cong \mathcal{O}(G) \otimes_{\mathcal{O}(H)} k \cong \frac{\mathcal{O}(G)}{\mathcal{O}(G)\, \varphi^\#(I_H)}.$$

Definition 1.19 (Identity component)**.** Let G be an algebraic group over k. The connected component of G containing the identity e is called the *identity component* of G, and is denoted by G°.

Proposition 1.20 (Largest finite étale subalgebra)**.** Let X be a finite-type k-scheme. Then there exists a largest finite étale k-subalgebra

$$\pi^0(X) \subseteq \mathcal{O}(X).$$

Proof. A finite étale k-algebra has the form

$$\prod_{i=1}^{r} L_i,$$

with each L_i/k a finite separable field extension. A quotient of a finite étale k-algebra is again finite étale.

If $A_1, A_2 \subseteq \mathcal{O}(X)$ are finite étale k-subalgebras, then their compositum inside $\mathcal{O}(X)$ is the image of

$$A_1 \otimes_k A_2 \longrightarrow \mathcal{O}(X),$$

hence is again finite étale. It therefore suffices to bound $\dim_k A$ for all finite étale k-subalgebras $A \subseteq \mathcal{O}(X)$.

Let k^s be a separable closure of k, and put $n = \dim_k A$. Then

$$A \otimes_k k^s \cong (k^s)^n.$$

Thus

$$1 = e_1 + \cdots + e_n,$$

where the e_i are pairwise orthogonal idempotents. Viewed as functions on X_{k^s}, these idempotents determine n pairwise disjoint nonempty open-and-closed subsets. Consequently,

$$n \leq \#\{\text{connected components of } X_{k^s}\},$$

which is finite because X is of finite type over k.

Choose a finite étale subalgebra of maximal possible k-dimension. Taking its compositum with any other finite étale subalgebra cannot increase the dimension, so it already contains every other one. This gives the largest such subalgebra $\pi^0(X)$. □

Definition 1.21 (The scheme of connected components)**.** Define

$$\pi_0(X) = \operatorname{Spec} \pi^0(X).$$

The inclusion

$$\pi^0(X) \hookrightarrow \mathcal{O}(X)$$

induces a canonical morphism

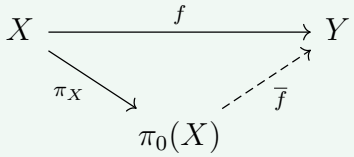
$$\pi_X : X \longrightarrow \pi_0(X).$$

The scheme $\pi_0(X)$ is finite étale over k.

Proposition 1.22 (Universal property)**.** The morphism $\pi_X : X \to \pi_0(X)$ is universal for morphisms from X to finite étale k-schemes. Namely, for every morphism $f : X \to Y$ with Y finite étale over k, there is a unique morphism $\overline{f} : \pi_0(X) \to Y$ such that

$$f = \overline{f} \circ \pi_X.$$

Equivalently,



commutes.

Proof. The map f induces

$$f^{\#} : \mathcal{O}(Y) \longrightarrow \mathcal{O}(X).$$

Its image is a quotient of a finite étale k-algebra, hence is finite étale and therefore lies in $\pi^0(X)$. This gives the factorization, and maximality of $\pi^0(X)$ gives uniqueness. □

Proposition 1.23 (Basic properties of π_0)**.** Let X, Y be finite-type k-schemes.

1. For every field extension k'/k,

$$\pi_0(X_{k'}) \cong \pi_0(X)_{k'}.$$

2.

$$\pi_0(X \times_k Y) \cong \pi_0(X) \times_k \pi_0(Y).$$

3. The fibers of $\pi_X : X \to \pi_0(X)$ are the connected components of X.

4. For every $x \in \pi_0(X)$, the fiber X_x is geometrically connected over $\kappa(x)$ (not

necessarily over k).

5. The morphism
$$X \longrightarrow \pi_0(X)$$
is faithfully flat.

Remark. See [Mil17, Propositions 1.30, 1.31, 1.33].

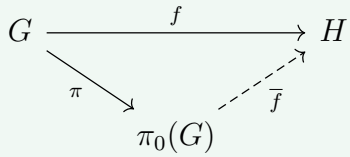
Corollary 1.24. Let X be a connected finite-type k-scheme such that $X(k) \neq \varnothing$. Then X is geometrically connected. Moreover, for every connected finite-type k-scheme Y, the product $X \times_k Y$ is connected.

Proof. Since X is connected,
$$\pi_0(X) = \operatorname{Spec} L$$
for some finite separable extension L/k. A k-rational point of X gives a composite
$$\operatorname{Spec} k \longrightarrow X \longrightarrow \operatorname{Spec} L,$$
hence a k-algebra map $L \to k$. Therefore $L = k$. By the geometric-connectedness property of the fibers, X is geometrically connected over k.

For connected Y,
$$\pi_0(X \times_k Y) \cong \pi_0(X) \times_k \pi_0(Y) = \operatorname{Spec} k \times_k \pi_0(Y) \cong \pi_0(Y),$$
which is single point. Hence $X \times_k Y$ is connected. □

Remark (The hypothesis $X(k) \neq \varnothing$ is essential)**.** Let
$$X = \operatorname{Spec} L,$$
where L/k is a finite Galois extension with $L \neq k$. Then X is connected, but
$$X_{\overline{k}} = \operatorname{Spec}(L \otimes_k \overline{k}) \cong \coprod_{[L:k]} \operatorname{Spec} \overline{k},$$
so X is not geometrically connected. Also,
$$X \times_k X = \operatorname{Spec}(L \otimes_k L) \cong \coprod_{[L:k]} \operatorname{Spec} L,$$
so $X \times_k X$ is not connected.

Proposition 1.25. Let G be an algebraic group over k. Then:

1. $\pi_0(G)$ is a finite étale algebraic group over k.
2. The canonical morphism
$$\pi : G \longrightarrow \pi_0(G)$$
is a homomorphism of algebraic groups, and it is universal among homomorphisms from G to finite étale algebraic groups. Thus, if H is finite étale and $f : G \to H$ is a homomorphism, there is a unique homomorphism $\overline{f} : \pi_0(G) \to H$ such that


commutes.
3. $$\ker\big(G \longrightarrow \pi_0(G)\big) = G^\circ.$$
In particular, G° is an algebraic subgroup of G.
4. For every field extension k'/k,
$$\pi_0(G_{k'}) \cong \pi_0(G)_{k'}, \qquad (G_{k'})^\circ \cong (G^\circ)_{k'}.$$
In particular, G° is geometrically connected.
5. The fibers of $G \to \pi_0(G)$ are the connected components of G.
6. For algebraic groups G, G',
$$\pi_0(G \times_k G') \cong \pi_0(G) \times_k \pi_0(G'), \qquad (G \times_k G')^\circ \cong G^\circ \times_k (G')^\circ.$$

Proof. For the multiplication, consider the composite

$$G \times G \xrightarrow{m_G} G \xrightarrow{\pi} \pi_0(G).$$

By the universal property of π_0, it factors uniquely through

$$\pi_0(G \times G) \cong \pi_0(G) \times \pi_0(G),$$

giving a multiplication

$$m_0 : \pi_0(G) \times \pi_0(G) \longrightarrow \pi_0(G)$$

such that

$$\begin{array}{ccc} G \times G & \xrightarrow{m_G} & G \\ {\scriptstyle \pi\times\pi}\downarrow & & \downarrow{\scriptstyle \pi} \\ \pi_0(G) \times \pi_0(G) & \xrightarrow[m_0]{} & \pi_0(G) \end{array}$$

commutes. The identity and inverse descend in the same way:

$$\operatorname{Spec} k \xrightarrow{e} G \xrightarrow{\pi} \pi_0(G), \qquad \begin{array}{ccc} G & \xrightarrow{\iota} & G \\ {\scriptstyle \pi}\downarrow & & \downarrow{\scriptstyle \pi} \\ \pi_0(G) & \xrightarrow[\iota_0]{} & \pi_0(G). \end{array}$$

These maps satisfy the group axioms, and π is a homomorphism. The universal property among homomorphisms follows from the universal property of π_0.

For (iii), the kernel is the fiber of π over the identity of $\pi_0(G)$; by the fiber description above, this is exactly the connected component G°.

For (iv), using the identity sections of the component groups,

$$\begin{aligned} (G_{k'})^\circ &= G_{k'} \times_{\pi_0(G_{k'})} \operatorname{Spec} k' \\ &\cong \left(G \times_{\pi_0(G)} \operatorname{Spec} k\right) \times_k k' \\ &= (\ker(G \to \pi_0(G)))_{k'} \\ &= (G^\circ)_{k'}. \end{aligned}$$

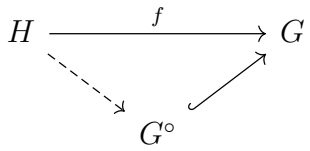
The remaining assertions follow from the basic properties of π_0. □

Definition 1.26 (Component group)**.** The finite étale algebraic group $\pi_0(G)$ is called the *component group*, or the *group of connected components*, of G.

Lemma 1.27. Let G be an algebraic group over k.

1. G is connected if and only if
$$\pi_0(G) = \operatorname{Spec} k.$$
2. If $f : H \to G$ is a homomorphism and H is connected, then f factors through

G°:



3. The identity component G° is geometrically connected, but the other connected components of G need not be geometrically connected over k.

Proof. For (1), a connected algebraic group has the k-rational identity point, so the preceding corollary gives geometric connectedness and forces its finite étale component scheme to be $\operatorname{Spec} k$. The converse is immediate from the description of the fibers of $G \to \pi_0(G)$.

For (2), the composite

$$H \longrightarrow G \longrightarrow \pi_0(G)$$

has connected image and contains the identity, so it is the identity morphism into the identity component of the finite étale scheme $\pi_0(G)$, namely the single point $\operatorname{Spec} k$. Hence $H \to G$ factors through the kernel G°.

Part (3) follows from base change for G°; the following example shows the second assertion. □

Example 1.28. Let $k = \mathbb{Q}$, and consider

$$\mu_3(R) = \{x \in R^\times : x^3 = 1\}, \qquad R \in \mathrm{Alg}_{\mathbb{Q}}.$$

Then

$$\mu_3 = \operatorname{Spec} \frac{\mathbb{Q}[T]}{(T^3-1)} \cong \operatorname{Spec} \mathbb{Q} \coprod \operatorname{Spec} \frac{\mathbb{Q}[T]}{(T^2+T+1)}.$$

The first component contains the identity. The second component has no $\mathbb{Q}$-point and is not geometrically connected: after base change to $\overline{\mathbb{Q}}$, it splits into two points. Thus connected components other than G° need not be geometrically connected over the ground field.

Definition 1.29 (Exact sequence of algebraic groups)**.** A sequence

$$e \longrightarrow N \xrightarrow{i} G \xrightarrow{q} Q \longrightarrow e$$

is called exact if:

1. i identifies N isomorphically with $\ker(q : G \to Q)$;

2. q is faithfully flat.

Example 1.30. The sequence

$$e \longrightarrow G^\circ \longrightarrow G \longrightarrow \pi_0(G) \longrightarrow e$$

is exact.

Proposition 1.31. Every connected algebraic group is irreducible. In fact, it is geometrically irreducible.

Proof. It is enough to prove that G is geometrically irreducible. Since a connected algebraic group is geometrically connected, after base change we may assume $k = \overline{k}$.

The reduced closed subscheme $G_{\mathrm{red}} \subseteq G$ has the same underlying topological space:

$$|G_{\mathrm{red}}| = |G|.$$

Since k is perfect, G_{red} is an algebraic subgroup of G, and a reduced algebraic group over a perfect field is smooth. Thus, for the topological question of irreducibility, we may replace G by the smooth reduced group G_{red}.

Suppose G had more than one irreducible component. Because G is connected, two irreducible components must meet. Choose a point x in such an intersection. Then the local ring $\mathcal{O}_{G,x}$ has at least two distinct minimal prime ideals. On the other hand, smoothness implies that $\mathcal{O}_{G,x}$ is a regular local ring, hence an integral domain, so it has only one minimal prime. This is a contradiction. □

Proposition 1.32. Let $\varphi : G \to H$ be a homomorphism of algebraic groups. Then the image $\varphi(|G|) \subseteq |H|$ is closed.

Proof. Step 1. It is enough to reduce to the case $k = \overline{k}$.

First, for any finite-type k-scheme X, the projection

$$p_X : |X_{\overline{k}}| \longrightarrow |X|$$

is a quotient map of topological spaces.

Second, if

$$p_H : |H_{\overline{k}}| \longrightarrow |H|$$

is the projection, then

$$\varphi_{\overline{k}}\big(|G_{\overline{k}}|\big) = p_H^{-1}\big(\varphi(|G|)\big).$$

Therefore $\varphi(|G|)$ is closed in $|H|$ if and only if the image after base change to $\overline{k}$ is closed in $|H_{\overline{k}}|$. Hence the proposition reduces to the algebraically closed case. [2]

Step 2. It is enough to work after passing to an algebraic closure, so assume $k = \overline{k}$. Then $|G| = G(k)$ on closed points. Put

$$N := \varphi(G(k)) \subseteq H(k),$$

and let $\overline{N}$ denote its closure in $|H|$. We show first that $\overline{N}$ is a subgroup of $H(k)$.

Let

$$i : H \xrightarrow{\sim} H, \qquad x \longmapsto x^{-1}.$$

Since i is a homeomorphism and $N^{-1} = N$,

$$(\overline{N})^{-1} = i(\overline{N}) = \overline{i(N)} = \overline{N}.$$

For $x \in N$, left translation

$$\ell_x : H \xrightarrow{\sim} H, \qquad y \longmapsto xy$$

satisfies

$$\ell_x(\overline{N}) = \overline{\ell_x(N)} = \overline{xN} = \overline{N}.$$

Hence $N\overline{N} \subseteq \overline{N}$. Now let $x \in \overline{N}$. Right translation

$$r_x : H \xrightarrow{\sim} H, \qquad y \longmapsto yx$$

gives

$$r_x(\overline{N}) = \overline{r_x(N)} = \overline{Nx} \subseteq \overline{N},$$

because $Nx \subseteq N\overline{N} \subseteq \overline{N}$. Thus

$$\overline{N}\,\overline{N} \subseteq \overline{N},$$

so $\overline{N}$ is a subgroup of $H(k)$.

Remark. This is the same idea as in the standard topological-group argument.

By Chevalley's theorem below, there exists a nonempty open dense subset

$$U \subseteq \overline{N}$$

[2] 2026.9.10 stopped at here.

with $U \subseteq N$. For any $x \in \overline{N}$, both U and $x^{-1}U$ are open dense subsets of $\overline{N}$. Hence

$$U \cap x^{-1}U \neq \varnothing.$$

Choose u in this intersection. Then $u \in U$ and $xu \in U$, so

$$x = (xu)u^{-1} \in UU^{-1} \subseteq N.$$

Therefore $\overline{N} \subseteq N$, hence $N = \overline{N}$ is closed. □

Theorem 1.33 (Chevalley)**.** Let $f : Y \to X$ be a finite-type morphism of noetherian schemes. Then $f(|Y|)$ is constructible in $|X|$. In particular, $f(|Y|)$ contains an open dense subset of its closure.

Corollary 1.34. If $\varphi : G \to H$ is a dominant homomorphism of algebraic groups, then φ is surjective on underlying topological spaces.

Proof. By the proposition, $\varphi(|G|)$ is closed in $|H|$. Dominance says that it is dense, hence it equals $|H|$. □

Proposition 1.35. Let $\varphi : G \to H$ be a surjective homomorphism of algebraic groups. If H is reduced, then φ is faithfully flat.

Proof. By generic flatness there is an open dense subset $U \subseteq H$ such that

$$\varphi^{-1}(U) \longrightarrow U$$

is flat. After extending scalars to an algebraic closure $\overline{k}$, the corresponding morphism over $U_{\overline{k}}$ is still flat.

For $g \in G(\overline{k})$, left translations give a commutative diagram

$$\begin{array}{ccc} G_{\overline{k}} & \xrightarrow[\sim]{L_g} & G_{\overline{k}} \\ {\scriptstyle \varphi_{\overline{k}}}\downarrow & & \downarrow{\scriptstyle \varphi_{\overline{k}}} \\ H_{\overline{k}} & \xrightarrow[\sim]{L_{\varphi(g)}} & H_{\overline{k}}. \end{array}$$

Thus $\varphi_{\overline{k}}$ is flat over every translate $\varphi(g)U_{\overline{k}}$. Surjectivity gives

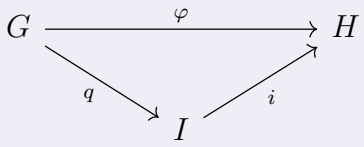
$$H_{\overline{k}} = \bigcup_{g \in G(\overline{k})} \varphi(g)U_{\overline{k}},$$

so $\varphi_{\overline{k}}$ is flat everywhere. Flatness descends along the faithfully flat extension $\overline{k}/k$, hence φ is flat. Since φ is surjective, it is faithfully flat. □

1.3 The image of a homomorphism

Motivation. For a homomorphism of abstract groups $G \xrightarrow{\varphi} H$, one factors φ through its image $\varphi(G)$. The analogous statement for algebraic groups is the following.

Theorem 1.36. Let $\varphi : G \to H$ be a homomorphism of algebraic groups, and suppose that G is smooth. Then φ factors as



where q is faithfully flat and i is a closed immersion.

Remark. The role of the smoothness hypothesis in this proof is to ensure that $G_{\overline{k}}$ is reduced; this forces the reduced image I to be geometrically reduced, which is what allows the group operations to restrict to I.

Proof. Let

$$I := \varphi(G)_{\mathrm{red}}$$

with its reduced induced closed-subscheme structure in H. Since G is reduced, φ factors through I; write the factorization as $G \xrightarrow{q} I \xrightarrow{i} H$. The map q is surjective, and I is reduced, so Proposition 1.35 shows that q is faithfully flat.

It remains to see that I is an algebraic subgroup of H. It is enough to prove that I is geometrically reduced. After base change to $\overline{k}$,

$$q_{\overline{k}} : G_{\overline{k}} \longrightarrow I_{\overline{k}}$$

is faithfully flat. Let $x \in |I_{\overline{k}}|$ and choose $g \in |G_{\overline{k}}|$ mapping to x. Then

$$\mathcal{O}_{I_{\overline{k}},x} \longrightarrow \mathcal{O}_{G_{\overline{k}},g}$$

is a faithfully flat local homomorphism, hence injective. Since G is smooth, $G_{\overline{k}}$ is reduced, so $\mathcal{O}_{G_{\overline{k}},g}$ is reduced. Therefore $\mathcal{O}_{I_{\overline{k}},x}$ is reduced. As x was arbitrary, $I_{\overline{k}}$ is reduced, hence I is geometrically reduced.

Now the multiplication and inverse maps of H carry the underlying sets $|I| \times |I|$ and $|I|$ into $|I|$. Since I and $I \times I$ are geometrically reduced, these morphisms factor through the

reduced closed subscheme I. Thus I inherits an algebraic-group structure and i is a closed immersion of algebraic groups. □

Definition 1.37. The algebraic subgroup I above is called the *image* of φ.

Remark. For linear algebraic groups, an image factorization of this form exists without assuming that G is smooth.

1.4 Group actions

Definition 1.38. An action of an algebraic group G on a k-scheme X is a morphism

$$\varphi : G \times_k X \longrightarrow X$$

such that the following two conditions hold.

1. If $e : \operatorname{Spec} k \to G$ is the identity section, then

$$\begin{array}{ccc} X = \operatorname{Spec} k \times X & \xrightarrow{e \times \mathrm{id}_X} & G \times X \\ & \searrow^{\mathrm{id}_X} & \downarrow{\scriptstyle \varphi} \\ & & X \end{array}$$

   commutes.

2. If $m : G \times G \to G$ is multiplication, then

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{m \times \mathrm{id}_X} & G \times X \\ {\scriptstyle \mathrm{id}_G \times \varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ G \times X & \xrightarrow[\varphi]{} & X \end{array}$$

   commutes.

Remark. By Yoneda's lemma, an action of G on X is equivalently a natural transformation

$$\widetilde{G} \times \widetilde{X} \longrightarrow \widetilde{X}$$

such that $G(R)$ acts on $X(R)$ for every k-algebra R.

Example 1.39. 1. The trivial action:

$$G \times X \xrightarrow{\mathrm{pr}_X} X, \qquad g \cdot x = x.$$

2. For $X = G$, left multiplication:

$$G \times G \xrightarrow{m} G, \qquad g \cdot x = gx.$$

3. For $X = G$, right multiplication written as a left action:

$$G \times G \longrightarrow G, \qquad g \cdot x = xg^{-1}.$$

4. For $X = G$, conjugation:

$$G \times G \longrightarrow G, \qquad g \cdot x = gxg^{-1}.$$

Definition 1.40. A linear representation of G is a homomorphism of algebraic groups

$$\rho : G \longrightarrow \mathrm{GL}_n$$

for some $n \geq 1$.

Remark. If $\rho : G \to \mathrm{GL}_n$ is a representation, then ρ gives an action of G on $\mathbb{A}^n$. On R-points,

$$G(R) \times \mathbb{A}^n(R) \longrightarrow \mathbb{A}^n(R), \qquad \left(g, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right) \longmapsto \rho(g) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

for every k-algebra R.

Lemma 1.41. Let $\varphi : G \times X \to X$ be an action. Then φ is faithfully flat. Moreover, φ is smooth if G is smooth over k, and φ is finite if G is finite over k.

Proof. The map

$$\alpha : G \times X \xrightarrow{\sim} G \times X, \qquad (g, x) \longmapsto (g, gx)$$

is an isomorphism, with inverse $(g, y) \mapsto (g, g^{-1}y)$. It satisfies

$$\varphi = \mathrm{pr}_X \circ \alpha.$$

Thus every property listed in the statement for φ is the corresponding property of the projection $\mathrm{pr}_X : G \times X \to X$, which is the base change of $G \to \operatorname{Spec} k$. □

Definition 1.42. Let $\varphi : G \times X \to X$ be an action and let $x \in X(k)$. The *orbit map* is

$$\varphi_x : G \simeq G \times \operatorname{Spec} k \xrightarrow{\mathrm{id}_G \times x} G \times X \xrightarrow{\varphi} X, \qquad g \longmapsto gx.$$

The orbit of x is the image of $|\varphi_x|$ and is denoted by $G \cdot x$.

Lemma 1.43. The orbit $G \cdot x$ is a locally closed subset of $|X|$.

Proof. As in the earlier reduction, it is enough to work after passing to an algebraic closure, so assume $k = \overline{k}$; the same reduction uses that $|X_{\overline{k}}| \to |X|$ is a quotient map. By Chevalley's theorem applied to $\varphi_x : G \to X$, there is a nonempty subset

$$U \subseteq G \cdot x$$

which is open and dense in $\overline{G \cdot x}$. Since U is nonempty and contained in the orbit,

$$G \cdot x = G \cdot U = \bigcup_{g \in G(k)} gU.$$

For every $g \in G(k)$, translation by g preserves $\overline{G \cdot x}$, and gU is open in $\overline{G \cdot x}$. Hence $G \cdot x$ is open in its closure, i.e. it is locally closed. □

Remark. It is possible that $X(k) = \varnothing$. For example, let

$$k = \mathbb{Q}, \qquad G = (\mathbb{Z}/2\mathbb{Z})_k, \qquad X = \operatorname{Spec} \mathbb{Q}(\sqrt{2}).$$

Then G acts on X by Galois conjugation, while $X(\mathbb{Q}) = \varnothing$.

Proposition 1.44. Assume $k = \overline{k}$ and that G acts on X. Every orbit of minimum dimension is closed. In particular, if $X \neq \varnothing$, a closed orbit exists.

Proof. For a subset $X_1 \subseteq X$, write

$$G \cdot X_1 := \{gx \in X : g \in G(k),\ x \in X_1\}.$$

First note that the closure of an orbit is G-stable, hence is a union of orbits. Indeed, for every $g \in G(k)$, the map

$$\ell_g : X \xrightarrow{\sim} X, \qquad x \longmapsto gx$$

is an isomorphism, and therefore

$$\ell_g(\overline{G \cdot x}) = \overline{\ell_g(G \cdot x)} = \overline{g(G \cdot x)} = \overline{G \cdot x}.$$

Thus $G \cdot \overline{G \cdot x} = \overline{G \cdot x}$.

Now let

$$y \in \overline{G \cdot x} \setminus G \cdot x.$$

Then

$$G \cdot y \subseteq \overline{G \cdot x} \setminus G \cdot x.$$

By Lemma 1.43, $G \cdot x$ is open and dense in $\overline{G \cdot x}$. Hence its boundary contains no irreducible component of $\overline{G \cdot x}$, and therefore has strictly smaller dimension. Since $G \cdot y$ is contained in the boundary,

$$\dim(G \cdot y) < \dim(G \cdot x).$$

Therefore an orbit of minimum dimension has empty boundary and is closed. □

1.5 Structure of general algebraic groups

Remark. Reference: [Mil17, Chapter 8].

Recall. For an algebraic group G over k, there is an exact sequence

$$e \longrightarrow G^\circ \longrightarrow G \longrightarrow \pi_0(G) \longrightarrow e,$$

where G° is the connected identity component and $\pi_0(G)$ is the finite étale component group.

Theorem 1.45 (Barsotti–Chevalley)**.** Let G be a connected algebraic group over a field k. Then there exists an exact sequence

$$e \longrightarrow N \longrightarrow G \longrightarrow Q \longrightarrow e,$$

where N is a connected affine normal algebraic subgroup and Q is an abelian variety. If, in addition, G is smooth and k is perfect, then N is the unique connected affine normal subgroup variety with abelian-variety quotient.

Remark. This is the pair of statements referred to [Mil17, Theorem 8.27 and 8.28].

Recall. An abelian variety is an algebraic group that is connected, smooth, and complete. In particular, every abelian variety is commutative.

Theorem 1.46. Every algebraic group G over k has a greatest (maximal) affine algebraic quotient

$$G \longrightarrow G_{\mathrm{aff}},$$

universal among homomorphisms from G to affine algebraic groups.[3]

Remark. More generally, every algebraic group G has a maximal affine quotient. Write

$$G_{\mathrm{aff}} := \operatorname{Spm} \mathcal{O}(G).$$

The first question is whether $\mathcal{O}(G)$ carries the expected Hopf algebra structure. The multiplication, identity, and inverse of G induce

$$m^{\#} : \mathcal{O}(G) \longrightarrow \mathcal{O}(G \times G) \cong \mathcal{O}(G) \otimes_k \mathcal{O}(G), \qquad e^{\#} : \mathcal{O}(G) \longrightarrow k, \qquad i^{\#} : \mathcal{O}(G) \longrightarrow \mathcal{O}(G),$$

and these maps make $\mathcal{O}(G)$ a Hopf k-algebra.

Theorem 1.47. Let G be an algebraic group over k. Then:

(i) $\mathcal{O}(G)$ is a finitely generated k-algebra. In particular, $G_{\mathrm{aff}} = \operatorname{Spm} \mathcal{O}(G)$ is a linear algebraic group.

(ii) The natural morphism

$$\phi : G \longrightarrow G_{\mathrm{aff}}$$

is universal among homomorphisms from G to linear algebraic groups: if $\phi' : G \to H$ is a homomorphism and H is a linear algebraic group, then there is a unique homomorphism $\psi : G_{\mathrm{aff}} \to H$ such that $\phi' = \psi \circ \phi$.

(iii) The morphism ϕ is faithfully flat.

(iv) The kernel $\ker(\phi)$ is anti-affine, i.e.

$$\mathcal{O}(\ker \phi) = k.$$

In particular, $\ker(\phi)$ is commutative.

Remark. Reference. [Mil17, Proposition 8.36, Corollary 8.13] and [Dem70, Exp. VI, Theorem 12.2].

[3] 2026.9.14 stopped at here.

Consequence. There is an exact sequence

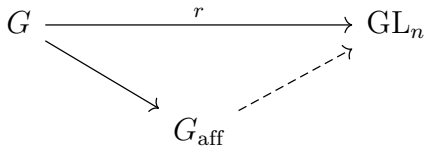
$$e \longrightarrow G_{\text{ant}} \longrightarrow G \longrightarrow G_{\text{aff}} \longrightarrow e,$$

where G_{ant} is anti-affine (hence commutative) and G_{aff} is a linear algebraic group.

Consequence. For every linear representation

$$r : G \longrightarrow \mathrm{GL}_n,$$

there is a unique factorization



Thus the study of linear representations of G is equivalent to the study of linear representations of G_{aff}. In what follows, we focus on linear algebraic groups.

2 Linear representations of linear algebraic groups

2.1 Hopf algebras

Recall. Let $G = \operatorname{Spm} A$ be a linear algebraic group over k. Then A has a Hopf algebra structure

$$m^{\#} : A \longrightarrow A \otimes_k A, \qquad e^{\#} : A \longrightarrow k, \qquad i^{\#} : A \longrightarrow A.$$

Geometrically, if

$$m^{\#}(f) = \sum_i g_i \otimes h_i,$$

then for every $R \in \mathrm{Alg}_k$ and $x, y \in G(R)$,

$$f(xy) = \sum_i g_i(x) h_i(y) \in R.$$

Moreover,

$$e^{\#}(f) = f(e), \qquad i^{\#}(f)(x) = f(x^{-1}).$$

Definition 2.1. Let A and B be Hopf k-algebras. A *homomorphism of Hopf algebras* $f : A \to B$ is a homomorphism of k-algebras such that

$$\begin{array}{ccc} A & \xrightarrow{\ f\ } & B \\ {\scriptstyle m_A^{\#}}\downarrow & & \downarrow{\scriptstyle m_B^{\#}} \\ A \otimes_k A & \xrightarrow[f \otimes f]{} & B \otimes_k B \end{array}$$

commutes.

Remark. For such an f, the counit and antipode are automatically compatible as well:

$$e_B^{\#} \circ f = e_A^{\#}, \qquad i_B^{\#} \circ f = f \circ i_A^{\#}.$$

Remark. Homomorphisms of linear algebraic groups

$$\operatorname{Spm} B \longrightarrow \operatorname{Spm} A$$

correspond contravariantly to homomorphisms of Hopf algebras

$$A \longrightarrow B.$$

Definition 2.2. Let B be a Hopf algebra over k. A k-subalgebra $A \subseteq B$ is called a *Hopf subalgebra* if

$$m_B^{\#}(A) \subseteq A \otimes_k A, \qquad i_B^{\#}(A) \subseteq A.$$

Proposition 2.3. Let $A \subseteq B$ be a Hopf subalgebra. Then A is a Hopf algebra with

$$m_A^{\#} = m_B^{\#}|_A, \qquad e_A^{\#} = e_B^{\#}|_A, \qquad i_A^{\#} = i_B^{\#}|_A.$$

Moreover, if $f : A \to B$ is a homomorphism of Hopf algebras, then $f(A)$ is a Hopf subalgebra of B.

Definition 2.4. Let A be a Hopf k-algebra. A *Hopf ideal* is an ideal $\mathfrak{a} \subseteq A$ such that

$$m^{\#}(\mathfrak{a}) \subseteq \mathfrak{a} \otimes_k A + A \otimes_k \mathfrak{a}, \qquad i^{\#}(\mathfrak{a}) \subseteq \mathfrak{a}, \qquad e^{\#}(\mathfrak{a}) = 0.$$

Equivalently, the composite

$$\mathfrak{a} \hookrightarrow A \xrightarrow{m^{\#}} A \otimes_k A \longrightarrow (A/\mathfrak{a}) \otimes_k (A/\mathfrak{a})$$

is zero, together with the antipode and counit conditions above.

Lemma 2.5. If $f : A \to B$ is a homomorphism of Hopf algebras, then $\ker f$ is a Hopf ideal of A.

Proof. Let $\mathfrak{a} = \ker f$. If $a \in \mathfrak{a}$, then

$$(f \otimes f)m_A^{\#}(a) = m_B^{\#}(f(a)) = 0.$$

Over the field k,

$$\ker(f \otimes f) = \mathfrak{a} \otimes_k A + A \otimes_k \mathfrak{a},$$

so

$$m_A^{\#}(\mathfrak{a}) \subseteq \mathfrak{a} \otimes_k A + A \otimes_k \mathfrak{a}.$$

The compatibility of f with the antipode and counit gives $i_A^{\#}(\mathfrak{a}) \subseteq \mathfrak{a}$ and $e_A^{\#}(\mathfrak{a}) = 0$. □

Proposition 2.6. Let $\mathfrak{a}$ be a Hopf ideal of a Hopf k-algebra A.

(i) The quotient $A/\mathfrak{a}$ has a Hopf algebra structure for which $A \to A/\mathfrak{a}$ is a homo-

morphism of Hopf algebras.

(ii) If $f : A \to B$ is a homomorphism of Hopf k-algebras, then the canonical map induces an isomorphism of Hopf algebras

$$A/\ker f \xrightarrow{\sim} f(A).$$

In particular, f factors as

$$A \twoheadrightarrow A/\ker f \xrightarrow{\sim} f(A) \hookrightarrow B.$$

Fact. Let $A \subseteq B$ be finitely generated Hopf k-algebras. Then B is faithfully flat over A. See [Mil17, Theorem 3.31].

Corollary 2.7. Every homomorphism

$$\varphi : G \longrightarrow H$$

of linear algebraic groups factors as

$$G \xrightarrow{q} I \xrightarrow{\iota} H,$$

where q is faithfully flat and ι is a closed immersion.

Corollary 2.8. Let $\varphi : G \to H$ be a homomorphism of linear algebraic groups. If

$$\ker \varphi = \operatorname{Spm} k,$$

then φ is a closed immersion.

Proof. By the preceding corollary, it is enough to treat the faithfully flat factor q. Thus we may assume that φ itself is faithfully flat, and it remains to show that φ is an isomorphism.

Since $\ker \varphi = \operatorname{Spm} k$, for every $R \in \operatorname{Alg}_k$ the map

$$\varphi(R) : G(R) \longrightarrow H(R)$$

is injective. We show that it is also surjective.

Take $y \in H(R)$. Since φ is faithfully flat, there is a faithfully flat R-algebra R' and an

element $x \in G(R')$ whose image in $H(R')$ is the pullback of y. The descent diagram is

$$\begin{array}{ccccc} G(R) & \longrightarrow & G(R') & \rightrightarrows & G(R' \otimes_R R') \\ {\scriptstyle \varphi(R)}\downarrow & & {\scriptstyle \varphi(R')}\downarrow & & \downarrow{\scriptstyle \varphi(R' \otimes_R R')} \\ H(R) & \longrightarrow & H(R') & \rightrightarrows & H(R' \otimes_R R'). \end{array}$$

The two pullbacks of x to $G(R' \otimes_R R')$ have the same image under $\varphi(R' \otimes_R R')$, because both come from y. Since $\varphi(R' \otimes_R R')$ is injective, the two pullbacks of x agree. Faithfully flat descent for the affine scheme G therefore gives a unique element of $G(R)$ descending to x, and its image under $\varphi(R)$ is y.

Thus $\varphi(R)$ is bijective for every $R \in \mathrm{Alg}_k$. By Yoneda, φ is an isomorphism. □

Remark. A faithfully flat homomorphism of linear algebraic groups $\varphi : G \to H$ need not be surjective on R-points for every $R \in \mathrm{Alg}_k$ when $\ker \varphi \neq \operatorname{Spm} k$.

Proof. Take $k = \mathbb{R}$, $G = H = \mathbb{G}_m$, and

$$\varphi : \mathbb{G}_m \longrightarrow \mathbb{G}_m, \qquad x \longmapsto x^2.$$

On coordinate rings,

$$\varphi^{\#} : \mathbb{R}[T, T^{-1}] \longrightarrow \mathbb{R}[T, T^{-1}], \qquad T \longmapsto T^2,$$

and

$$\varphi^{\#}\big(\mathbb{R}[T, T^{-1}]\big) = \mathbb{R}[T^2, T^{-2}].$$

The ring $\mathbb{R}[T, T^{-1}]$ is a free $\mathbb{R}[T^2, T^{-2}]$-module with basis $\{1, T\}$, so φ is faithfully flat. However,

$$\varphi(\mathbb{R}) : \mathbb{R}^\times \longrightarrow \mathbb{R}^\times, \qquad x \longmapsto x^2,$$

is not surjective. □

Let V be a k-vector space. Define the group-valued functor

$$\mathrm{GL}_V : \mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto \mathrm{GL}_V(R) := \mathrm{Aut}_R(V \otimes_k R).$$

If V is finite-dimensional, then after choosing a basis of V,

$$\mathrm{GL}_V \cong \mathrm{GL}_n, \qquad n = \dim_k V,$$

where the displayed isomorphism depends on the choice of basis.

2.2 Linear representations

Definition 2.9. Let G be a linear algebraic group over k and let V be a k-vector space. A *linear representation* of G on V is a homomorphism of algebraic groups

$$r : G \longrightarrow \mathrm{GL}_V .$$

The representation r is *faithful* if

$$\ker r = \mathrm{Spm}\, k,$$

and it is *trivial* if its image is the trivial group $\mathrm{Spm}\, k$.

Remark. For a representation $r : G \to \mathrm{GL}_V$, the following are equivalent:

r is faithful;
$r(R) : G(R) \longrightarrow \mathrm{GL}_V(R)$ is injective for every $R \in \mathrm{Alg}_k$;
r is a closed immersion.

Construction. Let $r : G \to \mathrm{GL}_V$ be a representation and write $G = \mathrm{Spm}\, A$. Since

$$G(A) = \mathrm{Hom}_{k\text{-alg}}(A, A),$$

the identity map id_A defines an A-point of G. Hence

$$r(A)(\mathrm{id}_A) \in \mathrm{Aut}_A(V \otimes_k A).$$

Define a k-linear map

$$\rho_r : V \longrightarrow V \otimes_k A, \qquad \rho_r(v) := r(A)(\mathrm{id}_A)(v \otimes 1).$$

Remark. Let $R \in \mathrm{Alg}_k$ and $x \in G(R)$, corresponding to a k-algebra map

$$x^{\#} : A \longrightarrow R.$$

Then naturality gives

$$\begin{array}{ccc} V & \xrightarrow{\ \rho_r\ } & V \otimes_k A \\ & \searrow^{v \mapsto r(R)(x)(v \otimes 1)} & \downarrow{\scriptstyle \mathrm{id}_V \otimes x^{\#}} \\ & & V \otimes_k R. \end{array}$$

Equivalently, the R-linearization of

$$(\mathrm{id}_V \otimes x^{\#}) \circ \rho_r : V \longrightarrow V \otimes_k R$$

is exactly $r(R)(x) : V \otimes_k R \to V \otimes_k R$.

2.3 Comodules

Definition 2.10. A *right $\mathcal{O}(G)$-comodule* is a k-vector space V together with a k-linear map

$$\rho : V \longrightarrow V \otimes_k \mathcal{O}(G)$$

such that

$$(\mathrm{id}_V \otimes e^{\#}) \circ \rho = \mathrm{id}_V$$

and

$$(\mathrm{id}_V \otimes m^{\#}) \circ \rho = (\rho \otimes \mathrm{id}_{\mathcal{O}(G)}) \circ \rho.$$

Proposition 2.11. The map ρ_r satisfies

$$(\mathrm{id}_V \otimes e^{\#}) \circ \rho_r = \mathrm{id}_V$$

and

$$(\rho_r \otimes \mathrm{id}_A) \circ \rho_r = (\mathrm{id}_V \otimes m^{\#}) \circ \rho_r.$$

Hence (V, ρ_r) is a right A-comodule.

Proof. The first identity is the coordinate-ring form of the identity action $e \cdot v = v$. The second is the coordinate-ring form of associativity of the action,

$$g_1\big(g_2 v\big) = (g_1 g_2) v.$$

Indeed, applying the construction to the universal points and using that r is a group homomorphism gives exactly the two displayed identities.[4] □

Definition 2.12. Let (V, ρ) be an $\mathcal{O}(G)$-comodule. An $\mathcal{O}(G)$-subcomodule of V is a

[4] 2026.9.17 stopped at here.

k-subspace $W \subseteq V$ such that

$$\rho(W) \subseteq W \otimes \mathcal{O}(G).$$

Then $(W, \rho|_W)$ is again an $\mathcal{O}(G)$-comodule.

Lemma 2.13. For a k-vector space V, there is a bijection

$$\left\{ \text{ representations } r : G \to \mathrm{GL}_V \right\} \longleftrightarrow \left\{ \mathcal{O}(G)\text{-comodule structures on } V \right\},$$

sending r to ρ_r.

Proof. It remains to define the inverse map. Let

$$\rho : V \longrightarrow V \otimes \mathcal{O}(G)$$

be a comodule map. For $R \in \mathrm{Alg}_k$ and

$$g \in G(R) = \mathrm{Hom}_{k\text{-alg}}(\mathcal{O}(G), R),$$

consider

$$V \xrightarrow{\rho} V \otimes \mathcal{O}(G) \xrightarrow{\mathrm{id}_V \otimes g} V \otimes R.$$

Its R-linearization is an element of

$$\mathrm{Aut}_R(V \otimes R) = \mathrm{GL}_V(R).$$

This defines $r_\rho(R) : G(R) \to \mathrm{GL}_V(R)$. The comodule identities give the identity and multiplication laws, and the two constructions are inverse to each other. □

Example 2.14 (The standard representation of GL_n)**.** Let $G = \mathrm{GL}_n$, let $V = k^n$ with basis $\{e_1, \ldots, e_n\}$, and let

$$r : \mathrm{GL}_n \xrightarrow{\sim} \mathrm{GL}_V$$

be the standard representation. Write

$$\mathcal{O}(G) = k[T_{ij}, \det(T)^{-1}].$$

If

$$\rho_r(e_i) = \sum_{j=1}^{n} e_j \otimes f_{ji},$$

then for $R \in \mathrm{Alg}_k$ and $g = (a_{st}) \in G(R)$,

$$\sum_{j=1}^{n} a_{ji} e_j = g e_i = \sum_{j=1}^{n} f_{ji}(g) e_j.^{5}$$

Hence $f_{ji}(g) = a_{ji}$, so $f_{ji} = T_{ji}$. Therefore

$$\rho_r(e_i) = \sum_{j=1}^{n} e_j \otimes T_{ji}.$$

Exercise 2.15. Compute the $\mathcal{O}(G)$-comodule structures on $V^\vee$ and on $V \otimes V$.

Proposition 2.16. Every $\mathcal{O}(G)$-comodule (V, ρ) is a filtered union of its finite-dimensional subcomodules.

Proof. It is enough to show that every $v \in V$ is contained in a finite-dimensional subcomodule. Let $\{f_i\}_{i \in I}$ be a k-basis of $\mathcal{O}(G)$. Write

$$\rho(v) = \sum_{i \in I_0} v_i \otimes f_i$$

for a finite subset $I_0 \subset I$, and let

$$W = \mathrm{span}_k\{v_i : i \in I_0\}.$$

The counit identity gives

$$v = (\mathrm{id}_V \otimes e^{\#})\rho(v) = \sum_{i \in I_0} v_i \, e^{\#}(f_i) \in W,$$

and $\dim_k W < \infty$.

[5] The second equality comes from another description of ρ_r which can be proved by $(\mathrm{id}_V \otimes g^{\#}) \circ \rho_r = r(R)(g)(- \circ 1)$.

It remains to show that $\rho(W) \subseteq W \otimes \mathcal{O}(G)$. For each $i \in I_0$, write

$$m^{\#}(f_i) = \sum_{j,\ell \in I} a_{ij\ell} f_j \otimes f_\ell,$$

where only finitely many coefficients occur. Coassociativity gives the commutative square

$$\begin{array}{ccc} V & \xrightarrow{\ \rho\ } & V \otimes \mathcal{O}(G) \\ {\scriptstyle \rho}\downarrow & & \downarrow{\scriptstyle \mathrm{id}_V \otimes m^{\#}} \\ V \otimes \mathcal{O}(G) & \xrightarrow[\rho \otimes \mathrm{id}]{} & V \otimes \mathcal{O}(G) \otimes \mathcal{O}(G). \end{array}$$

Applying the two paths to v, one obtains

$$\sum_i v_i \otimes m^{\#}(f_i) = \sum_{i,j,\ell} a_{ij\ell} v_i \otimes f_j \otimes f_\ell$$

and

$$\sum_\ell \rho(v_\ell) \otimes f_\ell.$$

Comparing coefficients of the basis elements f_ℓ in the third tensor factor gives

$$\rho(v_\ell) = \sum_{i,j} a_{ij\ell} v_i \otimes f_j \in W \otimes \mathcal{O}(G).$$

Thus W is a finite-dimensional subcomodule containing v. Since the sum of two finite-dimensional subcomodules is again a finite-dimensional subcomodule, these subcomodules form a filtered system. □

Corollary 2.17. Every representation of G is a filtered union of its finite-dimensional subrepresentations.

Proof. This is the preceding proposition under the representation–comodule correspondence. □

2.4 LAG is linear

Proposition 2.18 (Affine right actions and comodule algebras)**.** Let X be an affine k-scheme. A k-algebra homomorphism

$$\rho : \mathcal{O}(X) \longrightarrow \mathcal{O}(X) \otimes \mathcal{O}(G)$$

which makes $\mathcal{O}(X)$ an $\mathcal{O}(G)$-comodule corresponds contravariantly to a right action

$$a : X \times G \longrightarrow X.$$

Proof. The counit condition is the commutativity of

$$\begin{array}{ccc} \mathcal{O}(X) & \xrightarrow{\rho} & \mathcal{O}(X) \otimes \mathcal{O}(G) \\ & \searrow^{\mathrm{id}} & \downarrow{\scriptstyle \mathrm{id} \otimes e^{\#}} \\ & & \mathcal{O}(X), \end{array}$$

which is dual to the commutative triangle

$$\begin{array}{ccc} X \times G & \xrightarrow{a} & X \\ \uparrow{\scriptstyle (\mathrm{id}_X, e)} & \nearrow_{\mathrm{id}_X} & \\ X & & \end{array}$$

for the identity axiom of a right action. The coassociativity condition is

$$\begin{array}{ccc} \mathcal{O}(X) & \xrightarrow{\rho} & \mathcal{O}(X) \otimes \mathcal{O}(G) \\ \downarrow{\scriptstyle \rho} & & \downarrow{\scriptstyle \mathrm{id} \otimes m^{\#}} \\ \mathcal{O}(X) \otimes \mathcal{O}(G) & \xrightarrow[\rho \otimes \mathrm{id}]{} & \mathcal{O}(X) \otimes \mathcal{O}(G) \otimes \mathcal{O}(G), \end{array}$$

which is dual to

$$\begin{array}{ccc} X \times G \times G & \xrightarrow{a \times \mathrm{id}_G} & X \times G \\ \downarrow{\scriptstyle \mathrm{id}_X \times m} & & \downarrow{\scriptstyle a} \\ X \times G & \xrightarrow[a]{} & X. \end{array}$$

Thus the two sets of axioms are equivalent. □

Remark. If $r : G \to \mathrm{GL}_V$ is a representation, then for every $R \in \mathrm{Alg}_k$, the group $G(R)$ acts

on $V \otimes R$ on the left. The action extends to $\mathrm{Sym}(V)$, and because Spm is contravariant, G acts on $\operatorname{Spm}\mathrm{Sym}(V)$ on the right. Equivalently, the comodule map on V extends to a comodule-algebra map

$$\mathrm{Sym}(V) \longrightarrow \mathrm{Sym}(V) \otimes \mathcal{O}(G).$$

Definition 2.19. A representation V of G is *simple* if $V \neq 0$ and V has no nontrivial subrepresentations.

Corollary 2.20. Every simple representation of a linear algebraic group is finite-dimensional.

Proof. A nonzero vector is contained in a finite-dimensional subrepresentation by the preceding local-finiteness proposition. Simplicity forces that subrepresentation to be all of V. □

Let $X = G$, and let

$$m : G \times G \longrightarrow G$$

be right multiplication. The induced map

$$m^{\#} : \mathcal{O}(G) \longrightarrow \mathcal{O}(G) \otimes \mathcal{O}(G)$$

gives $\mathcal{O}(G)$ its regular $\mathcal{O}(G)$-comodule structure.

Definition 2.21. The representation

$$r_G : G \longrightarrow \mathrm{GL}_{\mathcal{O}(G)}$$

associated with the comodule $(\mathcal{O}(G), m^{\#})$ is called the *regular representation* of G.

Theorem 2.22. The regular representation of G has a faithful finite-dimensional subrepresentation. In particular, the regular representation is faithful.

Proof. Let $V \subseteq \mathcal{O}(G)$ be a finite-dimensional subcomodule. The subcomodule condition is expressed by the commutative diagram

$$\begin{array}{ccc} V & \hookrightarrow & \mathcal{O}(G) \\ \downarrow{\scriptstyle \rho} & & \downarrow{\scriptstyle m^{\#}} \\ V \otimes \mathcal{O}(G) & \hookrightarrow & \mathcal{O}(G) \otimes \mathcal{O}(G). \end{array}$$

Choose a basis $f_1, \dots, f_n$ of V. Write

$$m^{\#}(f_j) = \sum_{i=1}^{n} f_i \otimes b_{ij}, \qquad b_{ij} \in \mathcal{O}(G).$$

Then the corresponding representation

$$G \longrightarrow \mathrm{GL}_V \simeq \mathrm{GL}_n$$

has matrix $(b_{ij}(g))$ at g, and hence its comorphism is

$$\mathcal{O}(\mathrm{GL}_n) = k[T_{ij}, \det(T)^{-1}] \longrightarrow \mathcal{O}(G), \qquad T_{ij} \longmapsto b_{ij}.$$

This representation is faithful exactly when this map is surjective, that is, when the b_{ij} generate $\mathcal{O}(G)$ as a k-algebra.

Applying $e^{\#} \otimes \mathrm{id}$ to $m^{\#}(f_j) = \sum_i f_i \otimes b_{ij}$ gives

$$f_j = \sum_{i=1}^{n} f_i(e) b_{ij}.$$

Therefore it is enough to choose V so that

(1) V generates $\mathcal{O}(G)$ as a k-algebra;

(2) V is an $\mathcal{O}(G)$-subcomodule;

(3) $\dim_k V < \infty$.

Because $\mathcal{O}(G)$ is a finitely generated k-algebra, choose finitely many algebra generators. By local finiteness, each generator is contained in a finite-dimensional subcomodule. The sum of those finitely many subcomodules satisfies all three conditions. Hence the resulting $G \to \mathrm{GL}_V$ is faithful. □

Corollary 2.23. If G is a linear algebraic group, then G is isomorphic to an algebraic subgroup of GL_n for some n.

Proof. Take the faithful finite-dimensional subrepresentation furnished by the theorem. A faithful finite-dimensional representation of a linear algebraic group is a closed immersion. □

2.5 Constructing all finite-dimensional representations

Definition 2.24. For a k-vector space V, the comodule

$$V \otimes \mathcal{O}(G)$$

with coaction

$$\mathrm{id}_V \otimes m^{\#} : V \otimes \mathcal{O}(G) \longrightarrow V \otimes \mathcal{O}(G) \otimes \mathcal{O}(G)$$

is called the *free comodule on* V.

Remark. If $\dim_k V = n$ and a basis of V is fixed, then

$$V \otimes \mathcal{O}(G) \simeq \mathcal{O}(G)^n,$$

with the componentwise free-comodule structure.

Proposition 2.25. Every finite-dimensional $\mathcal{O}(G)$-comodule V is a subcomodule of

$$\left(\mathcal{O}(G)^n, (m^{\#})^n\right)$$

for $n = \dim_k V$.

Proof. The coaction

$$\rho : V \longrightarrow V \otimes \mathcal{O}(G)$$

is injective, because

$$(\mathrm{id}_V \otimes e^{\#}) \circ \rho = \mathrm{id}_V .$$

Moreover, the diagram

$$\begin{array}{ccc} V & \xrightarrow{\quad\rho\quad} & V \otimes \mathcal{O}(G) \\ {\scriptstyle\rho}\downarrow & & \downarrow{\scriptstyle \mathrm{id}_V \otimes m^{\#}} \\ V \otimes \mathcal{O}(G) & \xrightarrow[\rho\otimes\mathrm{id}]{} & V \otimes \mathcal{O}(G) \otimes \mathcal{O}(G) \end{array}$$

commutes. Hence ρ is a comodule morphism from V into the free comodule $V \otimes \mathcal{O}(G)$. After choosing a basis of V,

$$V \otimes \mathcal{O}(G) \simeq \mathcal{O}(G)^n.$$

□

Theorem 2.26. Let

$$r : G \longrightarrow \mathrm{GL}_V$$

be a faithful finite-dimensional representation of G. Then every finite-dimensional representation W of G can be constructed from V by taking tensor products, finite direct sums, duals, and subquotients.

Proof. Step 1: reduction to $W \subseteq \mathcal{O}(G)$. By the preceding proposition, we may embed

$$W \hookrightarrow \mathcal{O}(G)^n$$

for some n. Let

$$W_i = \mathrm{im}\Big(W \hookrightarrow \mathcal{O}(G)^n \xrightarrow{\mathrm{pr}_i} \mathcal{O}(G)\Big).$$

Then each W_i is a finite-dimensional subcomodule of $\mathcal{O}(G)$, and the map

$$W \longrightarrow \bigoplus_i W_i$$

is injective. Hence it is enough to treat finite-dimensional subcomodules $W \subseteq \mathcal{O}(G)$.
Step 2: reduction to $G = \mathrm{GL}_V$. Since r is faithful, its comorphism

$$r^{\#} : \mathcal{O}(\mathrm{GL}_V) \longrightarrow \mathcal{O}(G)$$

is surjective.

If

$$\widetilde{W} \subseteq \mathcal{O}(\mathrm{GL}_V)$$

is a finite-dimensional $\mathcal{O}(\mathrm{GL}_V)$-subcomodule, then the Hopf-algebra identity

$$m_G^{\#} \circ r^{\#} = (r^{\#} \otimes r^{\#}) \circ m_{\mathrm{GL}_V}^{\#}$$

shows that

$$r^{\#}(\widetilde{W}) \subseteq \mathcal{O}(G)$$

is an $\mathcal{O}(G)$-subcomodule. Equivalently, one has the commutative diagram

$$\begin{array}{ccccc}
\widetilde{W} & \hookrightarrow & \mathcal{O}(\mathrm{GL}_V) & \xrightarrow{m^{\#}_{\mathrm{GL}_V}} & \mathcal{O}(\mathrm{GL}_V)\otimes\mathcal{O}(\mathrm{GL}_V) \\
\downarrow{\scriptstyle r^{\#}|_{\widetilde{W}}} & & \downarrow{\scriptstyle r^{\#}} & & \downarrow{\scriptstyle r^{\#}\otimes r^{\#}} \\
r^{\#}(\widetilde{W}) & \hookrightarrow & \mathcal{O}(G) & \xrightarrow[m^{\#}_{G}]{} & \mathcal{O}(G)\otimes\mathcal{O}(G).
\end{array}$$

Conversely, given a finite-dimensional subcomodule $W \subseteq \mathcal{O}(G)$, choose lifts in $\mathcal{O}(\mathrm{GL}_V)$ of a basis of W. By local finiteness of the regular representation of GL_V, those lifts are contained in a finite-dimensional subcomodule

$$\widetilde{W} \subseteq \mathcal{O}(\mathrm{GL}_V)$$

such that

$$r^{\#}(\widetilde{W}) \supseteq W.$$

Thus it is enough to treat $G = \mathrm{GL}_V$ case. [6]

Step 3: the required finite-dimensional subrepresentation. It suffices to find a finite-dimensional subrepresentation

$$\widetilde{W} \subseteq \mathcal{O}(\mathrm{GL}(V))$$

such that:

(1) $\widetilde{W}$ is constructed from V by tensor products, finite direct sums, duals, and subquotients;

(2) $\widetilde{W}$ generates $\mathcal{O}(\mathrm{GL}(V))$ as a k-algebra.

Indeed, condition (2) gives a surjective homomorphism of comodule algebras

$$\mathrm{Sym}(\widetilde{W}) \longrightarrow \mathcal{O}(\mathrm{GL}(V)).$$

The algebra $\mathrm{Sym}(\widetilde{W})$ is the filtered union of the finite-dimensional representations

$$\bigoplus_{q=0}^{d} \mathrm{Sym}^q(\widetilde{W}), \qquad d \geq 0.$$

Each symmetric power is a quotient of a tensor power. Hence every finite-dimensional subrepresentation of $\mathcal{O}(\mathrm{GL}(V))$ is a subquotient of a representation constructed from V by

[6] 2026.9.21 stopped at here.

the permitted operations.

Step 4: construction of $\widetilde{W}$. Choose a basis $e_1, \dots, e_n$ of V. Then

$$\mathrm{GL}(V) \simeq \mathrm{GL}_n, \qquad \mathcal{O}(\mathrm{GL}_n) = k[T_{ij} \mid 1 \le i, j \le n, \det(T)^{-1}].$$

Set

$$\widetilde{W} = \left(\bigoplus_{1 \le i,j \le n} kT_{ij} \right) \oplus k \det(T)^{-1}.$$

The comultiplication is given by

$$m^{\#}(T_{ij}) = \sum_{\ell=1}^{n} T_{i\ell} \otimes T_{\ell j}, \qquad m^{\#}(\det(T)^{-1}) = \det(T)^{-1} \otimes \det(T)^{-1}.$$

For $1 \le i \le n$, let

$$V_i = \bigoplus_{j=1}^{n} kT_{ij}, \qquad V_0 = k \det(T)^{-1}.$$

Then

$$\widetilde{W} = V_0 \oplus V_1 \oplus \cdots \oplus V_n,$$

and all the V_i are subrepresentations of the regular representation. The standard coaction on V is

$$\rho \colon V \longrightarrow V \otimes \mathcal{O}(\mathrm{GL}(V)), \qquad e_j \longmapsto \sum_{i=1}^{n} e_i \otimes T_{ij}.$$

It follows that

$$V_i \simeq V \quad (1 \le i \le n), \qquad V_0 \simeq \left(\bigwedge^{n} V \right)^{\vee}.$$

Thus $\widetilde{W}$ has property (1), and its displayed basis shows that it generates $\mathcal{O}(\mathrm{GL}(V))$, giving property (2). □

2.6 Chevalley's theorem

Let $r \colon G \to \mathrm{GL}(V)$ be a finite-dimensional representation, and let $W \subseteq V$ be a k-linear subspace. For a k-algebra R, write $V_R = V \otimes_k R$ and $W_R = W \otimes_k R$.

Proposition 2.27. The functor

$$\mathrm{Stab}_G(W) \colon \mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto \{g \in G(R) \mid r_R(g)(W_R) = W_R\},$$

is represented by an algebraic subgroup of G.

Proof. Let $(e_j)_{j\in J}$ be a basis of W and extend it to a basis $(e_i)_{i\in I\sqcup J}$ of V. Write the coaction as

$$\rho(e_j) = \sum_{i\in I\sqcup J} e_i \otimes f_{ij}, \qquad f_{ij} \in \mathcal{O}(G).$$

For every $R \in \mathrm{Alg}_k$ and $g \in G(R)$, one has

$$\begin{aligned} g \in \mathrm{Stab}_G(W)(R) &\iff r_R(g)(W_R) = W_R \\ &\iff r_R(g)(e_j) \in W_R \quad \text{for every } j \in J \\ &\iff f_{ij}(g) = 0 \quad \text{for every } i \in I,\ j \in J. \end{aligned}$$

Here containment of $r_R(g)(W_R)$ in W_R is equivalent to equality: in the chosen basis, $r_R(g)$ is then block upper triangular, and its invertibility forces its diagonal blocks to be invertible. Therefore

$$\mathrm{Stab}_G(W) \simeq \mathrm{Spm}\left(\mathcal{O}(G)/(f_{ij} \mid i \in I,\ j \in J)\right),$$

which is a closed subgroup of G. □

Definition 2.28. The algebraic subgroup

$$G_W := \mathrm{Stab}_G(W)$$

is called the *stabilizer of W in G*.

Theorem 2.29 (Chevalley)**.** Every algebraic subgroup $H \subseteq G$ is the stabilizer of a one-dimensional subspace L in a finite-dimensional representation of G.

Remark. This theorem is one of the main ingredients in proving the existence of the quotient G/H as a scheme.

Proof. Step 1: realize H as the stabilizer of a finite-dimensional subspace. Let

$$\mathfrak{a} = \ker\big(\mathcal{O}(G) \longrightarrow \mathcal{O}(H)\big).$$

This is a Hopf ideal. Choose a finite-dimensional subcomodule $V \subseteq \mathcal{O}(G)$ containing a set of generators of $\mathfrak{a}$ as an ideal, and put

$$W = V \cap \mathfrak{a}.$$

Choose a basis $(f_j)_{j\in J}$ of W and extend it to a basis $(f_i)_{i\in I\sqcup J}$ of V. Write

$$m^{\#}(f_j) = \sum_{i\in I\sqcup J} f_i \otimes a_{ij}, \qquad j \in J,$$

and set

$$\mathfrak{a}' = (a_{ij} \mid i \in I,\ j \in J).$$

By the preceding proposition,

$$\mathrm{Stab}_G(W) = \mathrm{Spm}\big(\mathcal{O}(G)/\mathfrak{a}'\big).$$

Because $\mathfrak{a}$ is a Hopf ideal,

$$m^{\#}(\mathfrak{a}) \subseteq \mathfrak{a}\otimes\mathcal{O}(G) + \mathcal{O}(G)\otimes\mathfrak{a}.$$

The images of $(f_i)_{i\in I}$ in $\mathcal{O}(G)/\mathfrak{a}$ are linearly independent. Reducing the displayed expression for $m^{\#}(f_j)$ modulo

$$\mathfrak{a}\otimes\mathcal{O}(G) + \mathcal{O}(G)\otimes\mathfrak{a}$$

therefore gives $a_{ij} \in \mathfrak{a}$ for $i \in I$ and $j \in J$. Hence $\mathfrak{a}' \subseteq \mathfrak{a}$.

Conversely, applying $e^{\#}\otimes\mathrm{id}$ to $m^{\#}(f_j)$ gives

$$f_j = \sum_{i\in I\sqcup J} f_i(e)a_{ij} = \sum_{i\in I} f_i(e)a_{ij} \in \mathfrak{a}',$$

because $f_i(e) = 0$ for $i \in J$. Thus $W \subseteq \mathfrak{a}'$. Since W contains a generating set of $\mathfrak{a}$ as an ideal, we have $\mathfrak{a} \subseteq \mathfrak{a}'$. It follows that $\mathfrak{a} = \mathfrak{a}'$ and hence $H = \mathrm{Stab}_G(W)$.

Step 2: replace W by a line. Let $d = \dim_k W$, set

$$\widetilde{V} = \bigwedge^d V, \qquad L = \bigwedge^d W \subseteq \widetilde{V}.$$

We show that $\mathrm{Stab}_G(L) = H$. It is enough to prove that, for every k-algebra R and every $\alpha \in \mathrm{GL}(V_R)$,

$$\alpha(W_R) = W_R \quad\Longleftrightarrow\quad \left(\bigwedge^d \alpha\right)(L_R) = L_R.$$

Choose a basis $e_1, \dots, e_d$ of W and extend it to a basis of V; put $w = e_1 \wedge \cdots \wedge e_d$. The

forward implication is immediate. For the reverse implication, suppose

$$\left(\bigwedge^{d} \alpha\right)(w) = cw \qquad \text{for some } c \in R^{\times}.$$

Since

$$W_R = \{v \in V_R \mid w \wedge v = 0\},$$

every $v \in W_R$ satisfies

$$0 = \left(\bigwedge^{d+1} \alpha\right)(w \wedge v) = cw \wedge \alpha(v).$$

As c is a unit, $\alpha(v) \in W_R$. Applying the same argument to α^{-1} gives the reverse containment, so $\alpha(W_R) = W_R$. This completes the proof. □

2.7 Characters and eigenspaces

Definition 2.30. A *character* of a linear algebraic group G over k is a homomorphism

$$\chi \colon G \longrightarrow \mathbb{G}_m.$$

Remark. The character χ is determined by its comorphism

$$\chi^{\#} \colon k[T, T^{-1}] \longrightarrow \mathcal{O}(G),$$

which fits into the commutative diagram

$$\begin{array}{ccc} k[T,T^{-1}] & \xrightarrow{\quad\chi^{\#}\quad} & \mathcal{O}(G) \\ \downarrow{\scriptstyle m^{\#}} & & \downarrow{\scriptstyle m^{\#}} \\ k[T,T^{-1}] \otimes k[T,T^{-1}] & \xrightarrow[\chi^{\#}\otimes\chi^{\#}]{} & \mathcal{O}(G) \otimes \mathcal{O}(G). \end{array}$$

Since $m^{\#}(T) = T \otimes T$, one has

$$m^{\#}\big(\chi^{\#}(T)\big) = \chi^{\#}(T) \otimes \chi^{\#}(T).$$

An element $a \in \mathcal{O}(G)^{\times}$ satisfying

$$m^{\#}(a) = a \otimes a$$

is called *group-like*. There is a bijection

$$\{\text{characters of } G \text{ over } k\} \longleftrightarrow \{\text{group-like elements of } \mathcal{O}(G)\}, \qquad \chi \longmapsto a(\chi) := \chi^{\#}(T).$$

The trivial character corresponds to $1 \in \mathcal{O}(G)$. For every $R \in \mathrm{Alg}_k$, the induced map is

$$\chi(R)\colon G(R) \longrightarrow R^{\times}, \qquad g \longmapsto a(\chi)(g).$$

Thus characters of G are precisely its one-dimensional representations.

Example 2.31. The characters of $\mathbb{G}_m$ are

$$\chi_n\colon \mathbb{G}_m \longrightarrow \mathbb{G}_m, \qquad x \longmapsto x^n, \qquad n \in \mathbb{Z}.$$

Proposition 2.32. Distinct group-like elements of $\mathcal{O}(G)$ are linearly independent over k.

Proof. Suppose that $a_1, \ldots, a_n$ are distinct group-like elements satisfying a nontrivial linear relation

$$\sum_{i=1}^{n} c_i a_i = 0, \qquad c_i \in k,$$

and choose such a relation with n minimal. Then every c_i is nonzero. Since $a_1 \neq a_2$, there is a k-linear functional $\lambda\colon \mathcal{O}(G) \to k$ such that $\lambda(a_1) \neq \lambda(a_2)$. Applying $\lambda \otimes \mathrm{id}$ to the comultiplication of the relation gives

$$\sum_{i=1}^{n} c_i \lambda(a_i) a_i = 0.$$

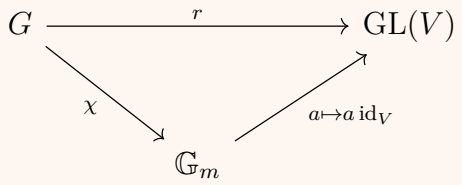
Subtracting $\lambda(a_1)$ times the original relation yields

$$\sum_{i=2}^{n} c_i \big(\lambda(a_i) - \lambda(a_1)\big) a_i = 0.$$

The coefficient of a_2 is nonzero, so this is a nontrivial relation involving fewer than n group-like elements, contradicting minimality. □

Corollary 2.33. Distinct characters $\chi_1, \ldots, \chi_n$ of G are linearly independent over k, when identified with the elements $a(\chi_1), \ldots, a(\chi_n) \in \mathcal{O}(G)$.

Definition 2.34. Let $r\colon G \to \mathrm{GL}(V)$ be a representation and let $\chi\colon G \to \mathbb{G}_m$ be a character. We say that G *acts on* V *through* χ if r factors as



Equivalently, the coaction satisfies

$$\rho(v) = v \otimes a(\chi) \qquad \text{for every } v \in V.$$

Definition 2.35. Let $r\colon G \to \mathrm{GL}(V)$ be a representation and let $W \subseteq V$ be a G-stable subspace. We say that G *acts on* W *through* χ if the restricted representation on W acts through χ. The largest subspace of V on which G acts through χ is called the χ-*eigenspace* (or χ-*weight space*) and is denoted by V_χ. Explicitly,

$$V_\chi = \{v \in V \mid \rho(v) = v \otimes a(\chi)\}^{7}.$$

Proposition 2.36. If V is a sum of character eigenspaces, then the sum is direct:

$$V = \sum_\chi V_\chi \qquad \Longrightarrow \qquad V = \bigoplus_\chi V_\chi.$$

Proof. Suppose that $v_1 + \cdots + v_s = 0$, with $v_i \in V_{\chi_i}$ and the χ_i pairwise distinct. Applying the coaction gives

$$0 = \sum_{i=1}^{s} \rho(v_i) = \sum_{i=1}^{s} v_i \otimes a(\chi_i).$$

For every $f \in V^\vee = \mathrm{Hom}_k(V, k)$, applying $f \otimes \mathrm{id}_A$ yields $\sum_i f(v_i)a(\chi_i) = 0$. The elements $a(\chi_i)$ are linearly independent, so $f(v_i) = 0$ for every f and every i. Hence all $v_i = 0$. □

Remark. If $G = \mathbb{G}_m^n$, every finite-dimensional representation of G is a direct sum of its character eigenspaces.

[7] 2026.9.24 stopped at here.

Example 2.37. Consider the two-dimensional representation of $\mathbb{G}_a$ given by

$$t \longmapsto \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

If χ_0 is the trivial character, then $V_{\chi_0} = ke_1$ is the only nonzero character eigenspace of this representation; it is one dimensional. In particular, the preceding corollary does not extend to arbitrary G.

2.8 Reconstructing the group from representations

Motivation. For a commutative group G, consider its group of continuous characters $G^\vee = \mathrm{Hom}_{\mathrm{cont}}(G, S^1)$. For locally compact Hausdorff abelian groups, Pontryagin duality gives $G^{\vee\vee} \simeq G$. This suggests the question: can a linear algebraic group be recovered from its representations? The answer is yes. A reference is Deligne–Milne, *Tannakian Categories*.

Definition 2.38. $\mathrm{Rep}_k(G)$ is the category of representations of G on finite-dimensional k-vector spaces. It is a tensor category: for $V, W \in \mathrm{Rep}_k(G)$, also $V \otimes_k W \in \mathrm{Rep}_k(G)$. The forgetful functor is

$$\omega : \mathrm{Rep}_k(G) \longrightarrow \mathrm{Vec}_k, \qquad (V, r) \longmapsto V,$$

where Vec_k denotes finite-dimensional k-vector spaces.

Definition 2.39. Define the group-valued functor

$$\mathrm{Aut}^{\otimes}(\omega) : \mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto \mathrm{Aut}^{\otimes}(\omega)(R).$$

An element $\lambda = (\lambda_X)_X$ of $\mathrm{Aut}^{\otimes}(\omega)(R)$ assigns to each $X \in \mathrm{Rep}_k(G)$ an R-linear automorphism

$$\lambda_X : \omega(X) \otimes_k R \xrightarrow{\sim} \omega(X) \otimes_k R$$

such that:

(1) $\lambda_{\mathbf{1}} = \mathrm{id}_R$ for the trivial representation $\mathbf{1}$;

(2) $\lambda_{X \otimes Y} = \lambda_X \otimes_R \lambda_Y$ for all X, Y;

(3) for every morphism $\alpha : X \to Y$ in $\mathrm{Rep}_k(G)$, the following diagram commutes:

$$\begin{array}{ccc} \omega(X)\otimes_k R & \xrightarrow{\lambda_X} & \omega(X)\otimes_k R \\ {\scriptstyle \omega(\alpha)\otimes 1}\downarrow & & \downarrow{\scriptstyle \omega(\alpha)\otimes 1} \\ \omega(Y)\otimes_k R & \xrightarrow[\lambda_Y]{} & \omega(Y)\otimes_k R. \end{array}$$

Multiplication in $\mathrm{Aut}^{\otimes}(\omega)(R)$ is componentwise composition.

Every $g \in G(R)$ gives an element of $\mathrm{Aut}^{\otimes}(\omega)(R)$: for $X = (V, r)$ take $\lambda_X = r_R(g)$ on $V \otimes_k R$. The three conditions above hold, giving a natural homomorphism of group-valued functors

$$G \longrightarrow \mathrm{Aut}^{\otimes}(\omega), \qquad g \longmapsto \big(r_R(g)\big)_{(V,r)}.$$

Theorem 2.40 (Reconstruction)**.** For a linear algebraic group G over k, the natural homomorphism

$$G \xrightarrow{\sim} \mathrm{Aut}^{\otimes}(\omega)$$

is an isomorphism of group-valued functors.

Remark. The corresponding statement for a general algebraic group cannot recover the entire group in this way; its finite-dimensional linear representations recover its affinization G^{aff}.

Let $R \in \mathrm{Alg}_k$. Put $G_R = G \times_k \operatorname{Spec} R$ and $A_R = \mathcal{O}(G_R) = A \otimes_k R$. The right regular representation is

$$r_{A,R'}(g)f(x) = f(xg) \quad (g, x \in G(R'),\ f \in A_R \otimes_R R'),$$

for each R-algebra R'. When $R = k$ we simply write r_A.

Lemma 2.41 (Translations)**.** Let $u : A_R \to A_R$ be an R-algebra endomorphism. If, with $\Delta_R = m^* : A_R \to A_R \otimes_R A_R$,

$$\begin{array}{ccc} A_R & \xrightarrow{\Delta_R} & A_R \otimes_R A_R \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \mathrm{id}\otimes u} \\ A_R & \xrightarrow[\Delta_R]{} & A_R \otimes_R A_R \end{array}$$

commutes, then $u = r_{A,R}(g)$ for some $g \in G(R)$.

Proof. Let $\varphi : G_R \to G_R$ correspond to u. For every R-algebra R', the displayed identity says $\varphi_{R'}(xy) = x\varphi_{R'}(y)$ for $x, y \in G(R')$. Taking $y = e$ and setting $g = \varphi_R(e) \in G(R)$ gives $\varphi_{R'}(x) = xg_{R'}$ for all R'. Hence u is right translation by g. □

Proof of the reconstruction theorem. Fix $R \in \mathrm{Alg}_k$. We prove that

$$G(R) \longrightarrow \mathrm{Aut}^{\otimes}(\omega)(R)$$

is bijective.

Step 1: Extend λ. Let $\lambda = (\lambda_X)_X \in \mathrm{Aut}^{\otimes}(\omega)(R)$. The conditions defining λ also apply to an infinite-dimensional rational representation, because every such representation is the directed union of its finite-dimensional subrepresentations. Naturality on their inclusions makes the maps λ_X compatible, so they define a map on the union. In particular λ acts on the regular representation A. For any rational representation $V = \bigcup_i V_i$ with finite-dimensional subrepresentations V_i, the resulting automorphism of $V \otimes_k R$ is characterized by

$$\lambda_V|_{V_i \otimes_k R} = \lambda_{V_i} \qquad \text{for all } i.$$

Its inverse is obtained by extending the inverses $\lambda_{V_i}^{-1}$.

Step 2: Recover a point from the regular representation. Set $u = \lambda_A : A_R \to A_R$. The unit $k \to A$ and multiplication $m : A \otimes_k A \to A$ are equivariant for the right regular action (with diagonal action on $A \otimes_k A$). Tensor compatibility and naturality give $u(1) = 1$ and $u(fh) = u(f)u(h)$. Thus u is an R-algebra endomorphism.

The comultiplication $\Delta = m^* : A \to A \otimes_k A$ is equivariant when the first factor on the right is given the *trivial* G-action and the second factor has the right regular action: indeed $\Delta(r_A(g)f) = (\mathrm{id} \otimes r_A(g))\Delta(f)$. A tensor automorphism of the forgetful functor acts as the identity on trivial representations. Naturality of Δ therefore gives

$$\begin{array}{ccc} A_R & \xrightarrow{\Delta_R} & A_R \otimes_R A_R \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \mathrm{id}\otimes u} \\ A_R & \xrightarrow[\Delta_R]{} & A_R \otimes_R A_R. \end{array}$$

By the translation lemma, $u = r_{A,R}(g)$ for some $g \in G(R)$.

Step 3: Control every representation. For $X = (V, r) \in \mathrm{Rep}_k(G)$, choose a basis of V and regard $A \otimes_k V$ as a finite direct sum of copies of the regular representation. The matrix-coefficient map

$$\iota : V \longrightarrow A \otimes_k V, \qquad \iota(v)(x) = r(x)v$$

is G-equivariant and is injective, since evaluation at e sends $\iota(v)$ to v. Thus V embeds in $A^{\oplus \dim V}$. Functoriality gives the commutative diagram

$$\begin{array}{ccc} V \otimes_k R & \xrightarrow{\iota\otimes 1} & A_R \otimes_R (V \otimes_k R) \\ {\scriptstyle \lambda_V}\downarrow & & \downarrow{\scriptstyle u\otimes \mathrm{id}} \\ V \otimes_k R & \xrightarrow[\iota\otimes 1]{} & A_R \otimes_R (V \otimes_k R). \end{array}$$

Since $u = r_{A,R}(g)$ and ι is equivariant, the same diagram commutes with $r_R(g)$ in place of λ_V. The top arrow remains injective after base change to R; hence $\lambda_V = r_R(g)$.

Step 4: Uniqueness. There is a finite-dimensional faithful representation $r : G \hookrightarrow \mathrm{GL}(V)$ which is a closed immersion. If two points of $G(R)$ give the same tensor automorphism, their images under r_R coincide, and so the points coincide. This proves bijectivity for every R, naturally in R. □

Remark. Reconstruction allows us to replace G by GL_n in some situations.

2.9 Characterization of categories of representations

Theorem 2.42 (Neutral Tannakian reconstruction)**.** Let $(\mathcal{C}, \otimes)$ be a neutral Tannakian category over k, and let $\omega : \mathcal{C} \to \mathrm{Vec}_k$ be a k-valued fibre functor. Then $\mathrm{Aut}^{\otimes}(\omega)$ is represented by an affine group scheme G over k, and the functor

$$\mathcal{C} \longrightarrow \mathrm{Rep}_k(G)$$

defined by ω is an equivalence of tensor categories.

References. Milne, *Tannakian Categories*, Section 9.47; Deligne–Milne, *Tannakian Categories*, Theorem 2.11.[8]

Definition 2.43 (Neutral Tannakian category)**.** A category $(\mathcal{C}, \otimes, \phi, \psi)$ is called a *neutral Tannakian category over k* if the following hold.

(1) $\mathcal{C}$ is a k-linear abelian category.

(2) $\otimes : \mathcal{C} \times \mathcal{C} \longrightarrow \mathcal{C}$ is a k-bilinear functor.

[8] 2026.9.28 stopped at here.

(3) There are functorial isomorphisms

$$\phi_{X,Y,Z} : X \otimes (Y \otimes Z) \xrightarrow{\sim} (X \otimes Y) \otimes Z, \qquad \psi_{X,Y} : X \otimes Y \xrightarrow{\sim} Y \otimes X,$$

for all $X, Y, Z \in \mathrm{Ob}(\mathcal{C})$, satisfying the usual associativity and symmetry coherence identities. There is an identity object $\mathbf{1}$, with compatible unit isomorphisms

$$\mathbf{1} \otimes X \simeq X \simeq X \otimes \mathbf{1}.$$

In particular, the functor $X \longmapsto \mathbf{1} \otimes X$ is an equivalence of categories.

(4) Every $X \in \mathrm{Ob}(\mathcal{C})$ has a dual $X^\vee$, equipped with the evaluation morphism

$$\mathrm{ev}_X : X^\vee \otimes X \longrightarrow \mathbf{1},$$

satisfying: for every $T \in \mathrm{Ob}(\mathcal{C})$, evaluation induces a natural isomorphism

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}}(T, X^\vee) &\xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(T \otimes X, \mathbf{1}), \\ \alpha &\longmapsto \left(T \otimes X \xrightarrow{\alpha \otimes \mathrm{id}_X} X^\vee \otimes X \xrightarrow{\mathrm{ev}_X} \mathbf{1} \right). \end{aligned}$$

(5) There is a k-valued fibre functor

$$\omega : (\mathcal{C}, \otimes) \longrightarrow \underline{\mathrm{Vect}}_k;$$

that is, an exact, faithful, k-linear functor preserving the tensor structure, including its associativity, symmetry, and unit constraints.

2.10 Jordan decomposition

Motivation: Jordan normal form

Let $x \in \mathrm{GL}_n(\overline{k})$. Its Jordan normal form is

$$x = g^{-1} \operatorname{diag}\big(B_{a_1}^{n_1}, \ldots, B_{a_t}^{n_t}\big) g, \qquad g \in \mathrm{GL}_n(\overline{k}),$$

where each $a_i \neq 0$ and

$$B_a^r = \begin{pmatrix} a & 1 & & 0 \\ & a & \ddots & \\ & & \ddots & 1 \\ 0 & & & a \end{pmatrix}_{r \times r}$$

is a Jordan block. Put

$$x_s = g^{-1}\operatorname{diag}\big(a_1 I_{n_1}, \dots, a_t I_{n_t}\big) g.$$

Then x_s is semisimple.

Put

$$x_u = g^{-1}\operatorname{diag}\big(a_1^{-1} B_{a_1}^{n_1}, \dots, a_t^{-1} B_{a_t}^{n_t}\big) g.$$

Then x_u is unipotent and

$$x = x_s x_u = x_u x_s.$$

This is the Jordan decomposition of x.

Remark. We now replace $\overline{k}$ by k and assume that k is perfect. The difficulty is that the eigenvalues of x may not lie in k, and how to replace GL_n by an arbitrary linear algebraic group G.

Let V be a finite-dimensional k-vector space and $\alpha \in \mathrm{End}_k(V)$.

$$P_\alpha(T) = \det(T \operatorname{id}_V - \alpha) \in k[T]$$

is the characteristic polynomial for α.

Definition 2.44 (Eigenvalues lying in k; primary spaces)**.** We say that the eigenvalues of α lie in k if $P_\alpha(T)$ splits over k, so that

$$P_\alpha(T) = \prod_{i=1}^{t} (T - a_i)^{m_i}, \qquad a_i \in k,$$

where the a_i are distinct. For an eigenvalue $a \in k$, its *primary space* is

$$V^a = \{v \in V \mid (\alpha - a \operatorname{id}_V)^N v = 0 \text{ for some } N > 0\}.$$

When the eigenvalues lie in k, the primary decomposition is

$$V = \bigoplus_{i=1}^{t} V^{a_i}.$$

Definition 2.45 (Semisimple and unipotent endomorphisms)**.** Let $\alpha \in \mathrm{End}_k(V)$.

(1) α is *semisimple* if $\alpha_{\overline{k}} \in \mathrm{End}_{\overline{k}}(V \otimes_k \overline{k})$ is diagonalizable.

(2) α is *unipotent* if $(\alpha - \operatorname{id}_V)^N = 0$ for some $N > 0$. Equivalently, all its eigenvalues

in $\overline{k}$ are equal to 1.

Theorem 2.46 (Multiplicative Jordan decomposition)**.** Let V be a finite-dimensional vector space over a perfect field k, and let $\alpha \in \mathrm{Aut}_k(V)$. There exist unique automorphisms α_s, α_u of V such that

(a) $\alpha = \alpha_s\alpha_u = \alpha_u\alpha_s$;

(b) α_s is semisimple and α_u is unipotent.

Moreover, $\alpha_s, \alpha_u \in k[\alpha]$: each is a polynomial in α with coefficients in k.

Proof. The assertion is immediate if $V = 0$, so assume $V \neq 0$.

Case 1: the eigenvalues of α lie in k. Write

$$P_\alpha(T) = \prod_{i=1}^{t}(T - a_i)^{m_i}, \qquad V = \bigoplus_{i=1}^{t} V^{a_i}.$$

Since α is invertible, every a_i is nonzero. Define

$$\alpha_s\big|_{V^{a_i}} = a_i \,\mathrm{id}_{V^{a_i}}, \qquad \alpha_u = \alpha\alpha_s^{-1}.$$

Then α_s is semisimple and commutes with α. On V^{a_i},

$$(\alpha_u - \mathrm{id}_V)\big|_{V^{a_i}} = a_i^{-1}(\alpha - a_i \,\mathrm{id}_V)\big|_{V^{a_i}},$$

which is nilpotent. Thus α_u is unipotent, and (a) and (b) hold.

Polynomiality. To obtain $Q(\alpha) = \alpha_s$, it is enough that

$$Q(T) \equiv a_i \pmod{(T - a_i)^{m_i}} \qquad \text{for every } i.$$

Indeed, $\alpha|_{V^{a_i}}$ has characteristic polynomial $(T-a_i)^{m_i}$, so Cayley–Hamilton gives the required equality on each primary space. The factors $(T - a_i)^{m_i}$ are pairwise coprime; the Chinese remainder theorem therefore supplies such a $Q \in k[T]$. Similarly, choose $\widetilde{Q} \in k[T]$ with

$$\widetilde{Q}(T) \equiv a_i^{-1} \pmod{(T - a_i)^{m_i}} \qquad \text{for every } i.$$

Then

$$\alpha_s = Q(\alpha), \qquad \alpha_s^{-1} = \widetilde{Q}(\alpha), \qquad \alpha_u = \alpha\widetilde{Q}(\alpha) \in k[\alpha].$$

Uniqueness. Suppose also that $\alpha = \beta_s\beta_u = \beta_u\beta_s$, where β_s is semisimple and β_u is unipotent. β_s and β_u commutes with α, hence with α_s and α_u, since the latter are polynomials in α. Consequently,

$$\beta_s^{-1}\alpha_s = \beta_u\alpha_u^{-1}.$$

The left-hand side is semisimple and the right-hand side is unipotent: commuting semisimple maps are simultaneously diagonalizable over $\overline{k}$, and commuting unipotent maps are simultaneously upper triangular with all diagonal entries equal to 1. An endomorphism which is both semisimple and unipotent is the identity. Thus $\alpha_s = \beta_s$ and $\alpha_u = \beta_u$.

Case 2: the general case. Because k is perfect, there is a finite Galois extension k'/k over which all eigenvalues of α lie. Choose a k-basis of V, and let $A \in \mathrm{GL}_n(k)$ be the matrix of α.

By Case 1, over k' there is a unique decomposition

$$A = A_sA_u = A_uA_s, \qquad A_s, A_u \in \mathrm{GL}_n(k'),$$

with A_s semisimple and A_u unipotent. For every $\sigma \in \mathrm{Gal}(k'/k)$,

$$A = \sigma(A) = \sigma(A_s)\sigma(A_u) = \sigma(A_u)\sigma(A_s).$$

This is another decomposition of the same kind. By uniqueness,

$$\sigma(A_s) = A_s, \qquad \sigma(A_u) = A_u.$$

Thus $A_s, A_u \in \mathrm{GL}_n(k)$, defining $\alpha_s, \alpha_u \in \mathrm{Aut}_k(V)$ satisfying (a) and (b). Uniqueness over k follows from uniqueness over k'.

Polynomiality over k. Let $\mu_A(T)$ be the minimal polynomial of A over k' and set $d = \deg \mu_A$. Choose a polynomial $Q \in k'[T]$ of least degree such that $A_s = Q(A)$. Reduction modulo μ_A gives $\deg Q < d$. Write

$$Q(T) = \sum_{i=0}^{d-1} c_iT^i, \qquad c_i \in k'.$$

For every $\sigma \in \mathrm{Gal}(k'/k)$, the equalities $\sigma(A) = A$ and $\sigma(A_s) = A_s$ give

$$0 = \sum_{i=0}^{d-1} (\sigma(c_i) - c_i)A^i.$$

By the definition of the minimal polynomial, $I, A, \ldots, A^{d-1}$ are linearly independent over k'. Hence $\sigma(c_i) = c_i$ for all i and all σ, so $c_i \in k$. Therefore $Q \in k[T]$ and $\alpha_s \in k[\alpha]$. The same

argument applies to A_u, giving $\alpha_u \in k[\alpha]$. □

Definition 2.47 (Semisimple and unipotent parts)**.** The automorphisms α_s and α_u are called the *semisimple part* and *unipotent part* of α. The factorization

$$\alpha = \alpha_s \alpha_u = \alpha_u \alpha_s$$

is its *multiplicative Jordan decomposition.*

Example 2.48 (A nonperfect field)**.** Suppose $\operatorname{char} k = 2$ and k is not perfect. Choose $a \in k \setminus k^2$, and put

$$x = \begin{pmatrix} 0 & 1 \\ a & 0 \end{pmatrix} \in \mathrm{GL}_2(k).$$

Its characteristic polynomial is

$$P_x(T) = T^2 - a = (T - \sqrt{a})^2.$$

Over $\overline{k}$, its Jordan decomposition is

$$\begin{pmatrix} 0 & 1 \\ a & 0 \end{pmatrix} = \underbrace{\begin{pmatrix} \sqrt{a} & 0 \\ 0 & \sqrt{a} \end{pmatrix}}_{\text{semisimple}} \underbrace{\begin{pmatrix} 0 & 1/\sqrt{a} \\ \sqrt{a} & 0 \end{pmatrix}}_{\text{unipotent}}.$$

The second factor U satisfies $(U - I)^2 = 0$ in characteristic 2. Neither factor belongs to $\mathrm{GL}_2(k)$. If x had a Jordan decomposition over k, base change to $\overline{k}$ would contradict uniqueness there. Thus the perfectness assumption in the theorem is necessary.

Proposition 2.49. Let V, W be finite-dimensional vector spaces over a perfect field k, and let $\alpha \in \mathrm{Aut}_k(V)$ and $\beta \in \mathrm{Aut}_k(W)$.

(1) *Subspaces and quotients.* If $W \subseteq V$ is α-stable, then W is stable under α_s and α_u, and

$$\alpha|_W = (\alpha_s|_W)(\alpha_u|_W) = (\alpha_u|_W)(\alpha_s|_W)$$

is the Jordan decomposition of $\alpha|_W$. Likewise, the automorphisms induced by α_s and α_u on V/W are the semisimple and unipotent parts of the automorphism induced by α.

(2) *Homomorphisms.* If a k-linear map $\varphi : W \longrightarrow V$ satisfies $\varphi\beta = \alpha\varphi$, then

$$\varphi\beta_s = \alpha_s\varphi, \qquad \varphi\beta_u = \alpha_u\varphi.$$

Equivalently, from the commutative square

$$\begin{array}{ccc} W & \xrightarrow{\varphi} & V \\ {\scriptstyle\beta}\downarrow & & \downarrow{\scriptstyle\alpha} \\ W & \xrightarrow{\varphi} & V \end{array}$$

we obtain the two commutative squares

$$\begin{array}{ccc} W & \xrightarrow{\varphi} & V \\ {\scriptstyle\beta_s}\downarrow & & \downarrow{\scriptstyle\alpha_s} \\ W & \xrightarrow{\varphi} & V \end{array} \qquad \begin{array}{ccc} W & \xrightarrow{\varphi} & V \\ {\scriptstyle\beta_u}\downarrow & & \downarrow{\scriptstyle\alpha_u} \\ W & \xrightarrow{\varphi} & V \end{array}$$

(3) *Tensor products.* For $\alpha \otimes \beta \in \mathrm{Aut}_k(V \otimes_k W)$,

$$(\alpha \otimes \beta)_s = \alpha_s \otimes \beta_s, \qquad (\alpha \otimes \beta)_u = \alpha_u \otimes \beta_u.$$

Proof. Semisimplicity, unipotence, and the identities in question can be checked after extending scalars. By uniqueness, extending scalars preserves Jordan decomposition. We may therefore assume that all eigenvalues of α and β lie in k.

(1) Since $\alpha_s, \alpha_u \in k[\alpha]$, every α-stable subspace W is stable under both factors. The restrictions remain semisimple and unipotent, respectively, and commute. Hence

$$\alpha|_W = (\alpha_s|_W)(\alpha_u|_W)$$

is its Jordan decomposition. The same reasoning applies to the induced automorphisms on V/W.

(2) Factor φ through its image:

$$W \xrightarrow{\varphi} \varphi(W) \hookrightarrow V.$$

Both $\ker\varphi$ and $\varphi(W)$ are stable under the corresponding automorphisms. The original

commutative diagram factors as

$$\begin{array}{ccccc} W & \xrightarrow{\varphi} & \varphi(W) & \hookrightarrow & V \\ \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \alpha|_{\varphi(W)}} & & \downarrow{\scriptstyle \alpha} \\ W & \xrightarrow{\varphi} & \varphi(W) & \hookrightarrow & V \end{array}$$

The surjective map identifies $W/\ker\varphi$ with $\varphi(W)$. Uniqueness of Jordan decomposition preserves its factors under this identification. Applying (1) to the quotient and the inclusion proves the two asserted identities.

(3) Write the primary decompositions as

$$V = \bigoplus_i V^{a_i}, \qquad W = \bigoplus_j W^{b_j}.$$

Then

$$V \otimes_k W = \bigoplus_{i,j} V^{a_i} \otimes_k W^{b_j},$$

and $\alpha_s \otimes \beta_s$ acts on $V^{a_i} \otimes_k W^{b_j}$ by scalar multiplication by $a_i b_j$. In particular, it is semisimple. The automorphism $\alpha_u \otimes \beta_u$ is unipotent: in bases where α_u and β_u are upper triangular with diagonal entries 1, their tensor product has the same property. The two factors commute, and

$$\alpha \otimes \beta = (\alpha_s \otimes \beta_s)(\alpha_u \otimes \beta_u) = (\alpha_u \otimes \beta_u)(\alpha_s \otimes \beta_s).$$

Uniqueness gives the claimed semisimple and unipotent parts.[9] □

Definition 2.50. Let V be a k-vector space and let $\alpha\colon V \xrightarrow{\sim} V$ be a k-linear automorphism.

1. We say that α is *locally finite* if V is a union of finite-dimensional α-stable subspaces.
2. Suppose that α is locally finite. We say that α is *semisimple* (respectively, *locally unipotent*) if, for every finite-dimensional α-stable subspace $W \subseteq V$, the restriction $\alpha|_W$ is semisimple (respectively, unipotent). Here semisimple means diagonalizable after extension to an algebraic closure.

[9] 2026.10.8 stopped at here.

Remark. Local unipotence does not require a single integer $N > 0$ such that $(\alpha - \mathrm{id}_V)^N = 0$ on all of V. The nilpotence exponent may depend on the finite-dimensional stable subspace; there need not be a uniform upper bound.

Example 2.51. Let G be a linear algebraic group and let $r_V \colon G \to \mathrm{GL}_V$ be a rational representation. For every $g \in G(k)$, the automorphism $r_V(g) \in \mathrm{Aut}_k(V)$ is locally finite, since V is a union of finite-dimensional G-stable subspaces.

Corollary 2.52 (Locally finite Jordan decomposition)**.** Let $\alpha \in \mathrm{Aut}_k(V)$ be locally finite. There are unique locally finite automorphisms α_s, α_u of V, called the *semisimple part* and the *locally unipotent part*, such that

1. $\alpha = \alpha_s \alpha_u = \alpha_u \alpha_s$;
2. α_s is semisimple and α_u is locally unipotent.

This factorization is the *Jordan decomposition* of α.

Proof. Write $V = \bigcup_i W_i$, where the W_i are finite-dimensional α-stable subspaces, and define

$$\alpha_s|_{W_i} := (\alpha|_{W_i})_s, \qquad \alpha_u|_{W_i} := (\alpha|_{W_i})_u.$$

Claim. These automorphisms are well defined and glue together. Indeed, compatibility of finite-dimensional Jordan decomposition with restriction gives

$$(\alpha|_{W_i})_s|_{W_i \cap W_j} = (\alpha|_{W_i \cap W_j})_s = (\alpha|_{W_j})_s\big|_{W_i \cap W_j},$$

and the same equality holds for the unipotent parts. The resulting maps are linear: any two vectors lie in the finite-dimensional stable subspace $W_i + W_j$. Their inverses glue in the same way, so they are automorphisms. The factorization and commutativity can be checked on each W_i. Moreover, every finite-dimensional α_s-stable subspace is contained in some finite-dimensional α-stable subspace, on which α_s is semisimple. The analogous assertion holds for α_u.

For uniqueness, suppose $\alpha = \beta_s \beta_u = \beta_u \beta_s$ is another such decomposition. Every vector lies in a common finite-dimensional stable subspace for the commuting locally finite maps β_s, β_u: take its β_s-stable span and then the β_u-stable span of a basis. This subspace is also α-stable, so finite-dimensional uniqueness gives $\beta_s = \alpha_s$ and $\beta_u = \alpha_u$ on it, hence on V. $\square$

Theorem 2.53 (Jordan decomposition in a linear algebraic group)**.** Let G be a linear algebraic group over a perfect field k, and let $g \in G(k)$. There are unique elements $g_s, g_u \in G(k)$ such that

1. $g = g_s g_u = g_u g_s$;
2. for every $(V, r_V) \in \mathrm{Rep}_k(G)$,

$$r_V(g_s) = r_V(g)_s, \qquad r_V(g_u) = r_V(g)_u.$$

Here $\mathrm{Rep}_k(G)$ denotes the category of finite-dimensional rational representations of G.

Definition 2.54.
1. The elements g_s and g_u are the *semisimple part* and the *unipotent part* of g. The factorization $g = g_s g_u$ is the *Jordan decomposition* of g.
2. An element $g \in G(k)$ is *semisimple* if $g = g_s$, and *unipotent* if $g = g_u$.

Remark (The fibre functor and tensor automorphisms)**.** Recall the fibre functor

$$\omega \colon \mathrm{Rep}_k(G) \longrightarrow \mathrm{Vect}_k, \qquad (V, r_V) \longmapsto V.$$

The group-valued functor $\mathrm{Aut}^{\otimes}(\omega) \colon \mathrm{Alg}_k \to \mathrm{Grp}$ assigns to a k-algebra R the families $\lambda = (\lambda_V)_{(V,r_V)}$ of R-linear automorphisms

$$\lambda_V \colon V \otimes_k R \xrightarrow{\sim} V \otimes_k R$$

satisfying the following properties:

1. $\lambda_{\mathbf{1}} = \mathrm{id}_R$, where $\mathbf{1}$ is the trivial one-dimensional representation;
2. $\lambda_{V \otimes W} = \lambda_V \otimes_R \lambda_W$;
3. for every homomorphism of representations $\varphi \colon V \to W$, the following diagram commutes:

$$\begin{array}{ccc} V \otimes_k R & \xrightarrow{\varphi \otimes 1} & W \otimes_k R \\ {\scriptstyle \lambda_V}\downarrow & & \downarrow{\scriptstyle \lambda_W} \\ V \otimes_k R & \xrightarrow[\varphi \otimes 1]{} & W \otimes_k R. \end{array}$$

Remark (Reconstruction)**.** The natural map

$$G \xrightarrow{\sim} \mathrm{Aut}^{\otimes}(\omega)$$

is an isomorphism of group-valued functors on Alg_k. In particular,

$$G(k) \xrightarrow{\sim} \mathrm{Aut}^{\otimes}(\omega)(k), \qquad g \longmapsto \big(r_V(g)\big)_{(V,r_V)}.$$

Proof of the theorem. For every $(V, r_V) \in \mathrm{Rep}_k(G)$, form $r_V(g)_s, r_V(g)_u \in \mathrm{Aut}_k(V)$. *Claim.* The families

$$\big(r_V(g)_s\big)_{(V,r_V)}, \qquad \big(r_V(g)_u\big)_{(V,r_V)}$$

belong to $\mathrm{Aut}^{\otimes}(\omega)(k)$.

We verify the three properties above.

1. Since $r_{\mathbf{1}}(g) = \mathrm{id}_k$, both its semisimple and unipotent parts are id_k.

2. Compatibility of Jordan decomposition with tensor products gives

$$r_{V\otimes W}(g)_s = \big(r_V(g) \otimes r_W(g)\big)_s = r_V(g)_s \otimes r_W(g)_s,$$
$$r_{V\otimes W}(g)_u = \big(r_V(g) \otimes r_W(g)\big)_u = r_V(g)_u \otimes r_W(g)_u.$$

3. If $\varphi\colon V \to W$ is a homomorphism of representations, the square

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ {\scriptstyle r_V(g)}\downarrow & & \downarrow{\scriptstyle r_W(g)} \\ V & \xrightarrow[\varphi]{} & W \end{array}$$

commutes. Compatibility of Jordan decomposition with homomorphisms gives the two commutative squares

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ {\scriptstyle r_V(g)_s}\downarrow & & \downarrow{\scriptstyle r_W(g)_s} \\ V & \xrightarrow[\varphi]{} & W \end{array} \qquad \begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ {\scriptstyle r_V(g)_u}\downarrow & & \downarrow{\scriptstyle r_W(g)_u} \\ V & \xrightarrow[\varphi]{} & W. \end{array}$$

The claim follows. Reconstruction therefore supplies unique $g_s, g_u \in G(k)$ with the required identities in every representation.

For the factorization and commutativity, take a faithful representation $r_V\colon G \hookrightarrow \mathrm{GL}_V$. Then

$$r_V(g_s g_u) = r_V(g)_s r_V(g)_u = r_V(g),$$
$$r_V(g_u g_s) = r_V(g)_u r_V(g)_s = r_V(g).$$

Faithfulness gives $g = g_s g_u = g_u g_s$. Uniqueness also follows from this faithful representation and uniqueness of the Jordan decomposition in GL_V. □

Corollary 2.55. Let $\phi\colon G \to H$ be a homomorphism of linear algebraic groups over a perfect field k. For every $g \in G(k)$,

$$\phi(g_s) = \phi(g)_s, \qquad \phi(g_u) = \phi(g)_u.$$

Proof. Choose a faithful representation $r_V\colon H \hookrightarrow \mathrm{GL}_V$. The composite $(V, r_V \circ \phi)$ is a representation of G.

Applying the theorem to this composite representation and to H gives

$$\begin{aligned} r_V\big(\phi(g_s)\big) &= (r_V \circ \phi)(g_s) = \big((r_V \circ \phi)(g)\big)_s = r_V\big(\phi(g)_s\big), \\ r_V\big(\phi(g_u)\big) &= (r_V \circ \phi)(g_u) = \big((r_V \circ \phi)(g)\big)_u = r_V\big(\phi(g)_u\big). \end{aligned}$$

The conclusion follows from faithfulness of r_V. □

3 Quotient of Linear Algbraic Groups

3.1 Preliminaries

3.1.1 Density of points

Reference: [Mil17, Chapter I, §§1.9–1.18, pp. 9–11]. In this subsection, X is a scheme of finite type over a field k. Write $|X|$ for its underlying topological space.

Definition 3.1 (Schematic density)**.** Let $S \subseteq X(k)$. We say that S is *schematically dense* in X if the only closed subscheme $Z \subseteq X$ such that $S \subseteq Z(k)$ is X itself. In particular, schematic density of S implies that X is reduced.

Remark (The affine criterion)**.** Let $X = \operatorname{Spec} A$ and identify $X(k)$ with the maximal ideals $\mathfrak{m} \subset A$ whose residue field is k. For a closed subscheme $Z = \operatorname{Spec}(A/\mathfrak{a})$,

$$S \subseteq Z(k) \quad \Longleftrightarrow \quad \mathfrak{a} \subseteq \mathfrak{m} \text{ for every } \mathfrak{m} \in S.$$

Consequently,

$$S \text{ is schematically dense in } X \quad \Longleftrightarrow \quad \bigcap_{\mathfrak{m} \in S} \mathfrak{m} = 0.$$

Proposition 3.2. Let $S \subseteq X(k) \subseteq |X|$. The following are equivalent:

1. S is schematically dense in X;
2. X is reduced and S is dense in $|X|$;
3. writing $i_s \colon \operatorname{Spec} k \to X$ for each $s \in S$, the homomorphism of sheaves

$$\mathcal{O}_X \longrightarrow \prod_{s \in S} (i_s)_* \mathcal{O}_{\operatorname{Spec} k}$$

is injective. Equivalently, for every open subset $U \subseteq X$,

$$f \in \mathcal{O}_X(U), \qquad f(s) = 0 \text{ for all } s \in S \cap U \quad \Longrightarrow \quad f = 0.$$

In particular, schematic density is a local property on X.

Proposition 3.3. 1. A schematically dense subset $S \subseteq X(k)$ remains schematically dense after extension of the base field: for every field extension k'/k, its image in

$X_{k'}(k')$ is schematically dense in $X_{k'}$. This implies X is geometrically reduced.

2. In particular, if $X(k)$ is schematically dense in X, then X is geometrically reduced.

Remark. The converse of (2) does not hold. For example,

$$k = \mathbb{R}, \qquad X = \operatorname{Spec}\big(\mathbb{R}[x,y]/(x^2+y^2+1)\big)$$

is geometrically reduced but has no real points. Also, for $k = \mathbb{F}_p$ and $X = \mathbb{A}^1_k$, all k-points lie in the proper closed subscheme

$$Z = \operatorname{Spec}\big(k[T]/(T^p - T)\big) \subsetneq X.$$

Thus $X(k)$ is not schematically dense in X.

Definition 3.4 (Density of points over an extension)**.** Let k'/k be a field extension. We say that $X(k')$ is *schematically dense in* X if the only closed subscheme $Z \subseteq X$ such that $Z(k') = X(k')$ is X itself.

Proposition 3.5. Let k^s be a separable closure of k. If X is geometrically reduced, then $X(k^s)$ is schematically dense in X.

Example 3.6. If X is nonreduced, then $X(k^s)$ is not schematically dense in X: the proper closed subscheme X_{red} has the same k^s-points as X.

3.1.2 Frobenius

In this subsection, k is a field of characteristic $p > 0$.

Definition 3.7 (Absolute and relative Frobenius)**.** For every $R \in \mathrm{Alg}_k$, the map

$$f_R \colon R \longrightarrow R, \qquad x \longmapsto x^p$$

is a ring homomorphism. It need not be a homomorphism of k-algebras. Write $f = f_k$ and $\sigma_k \colon \operatorname{Spec} k \to \operatorname{Spec} k$ for the corresponding morphism.
For a scheme X over k, its *absolute Frobenius* $\sigma_X \colon X \to X$ is the identity on $|X|$ and

sends local sections a of $\mathcal{O}_X$ to a^p. If $q\colon X \to \operatorname{Spec} k$ is the structure morphism,

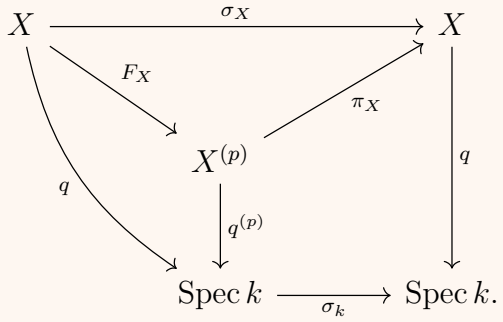
$$\begin{array}{ccc} X & \xrightarrow{\sigma_X} & X \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle q} \\ \operatorname{Spec} k & \xrightarrow[\sigma_k]{} & \operatorname{Spec} k \end{array}$$

commutes.

The *Frobenius twist* is

$$X^{(p)} := X \times_{\operatorname{Spec} k,\, \sigma_k} \operatorname{Spec} k.$$

The *relative Frobenius* $F_X\colon X \to X^{(p)}$ is the unique k-morphism in the diagram



Here the square formed by $X^{(p)}$, X, and the two copies of $\operatorname{Spec} k$ is cartesian, and $\pi_X F_X = \sigma_X$.

For $R \in \operatorname{Alg}_k$, let ${}_fR$ be R as a ring, with k-algebra structure

$$k \xrightarrow{f} k \longrightarrow R.$$

Then, naturally in R,

$$X^{(p)}(R) = X({}_fR).$$

The map

$$R \longrightarrow {}_fR, \qquad x \longmapsto x^p$$

is a homomorphism of k-algebras, and it induces

$$F_X(R)\colon X(R) \longrightarrow X^{(p)}(R) = X({}_fR).$$

For $X = \operatorname{Spec} A$, this corresponds to

$$F_X^{\#}\colon A \otimes_{k,f} k \longrightarrow A, \qquad a \otimes c \longmapsto c\, a^p.$$

Proposition 3.8 (Basic facts about relative Frobenius)**.**

1. *Functoriality.* For every k-morphism $\varphi\colon X \to Y$, the diagram

$$\begin{array}{ccc} X & \xrightarrow{\varphi} & Y \\ {\scriptstyle F_X}\downarrow & & \downarrow{\scriptstyle F_Y} \\ X^{(p)} & \xrightarrow[\varphi^{(p)}]{} & Y^{(p)} \end{array}$$

   commutes.

2. *Products.* Under the canonical isomorphism $(X \times_k Y)^{(p)} \simeq X^{(p)} \times_k Y^{(p)}$,

$$F_{X\times_k Y} = F_X \times_k F_Y.$$

3. *Base change.* For every field extension k'/k, the canonical identification

$$(X^{(p)})_{k'} \simeq (X_{k'})^{(p)}$$

   identifies $(F_X)_{k'}$ with $F_{X_{k'}}$.

Remark. If G is an algebraic group over k, then $G^{(p)}$ is an algebraic group over k: it represents the group-valued functor

$$\mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto G({}_fR) = G^{(p)}(R).$$

Proposition 3.9. The relative Frobenius $F_G\colon G \to G^{(p)}$ is a homomorphism of algebraic groups.

Proof. Functoriality and compatibility with products give the commutative diagram

$$\begin{array}{ccc} G \times_k G & \xrightarrow{m} & G \\ {\scriptstyle F_G\times F_G}\downarrow & & \downarrow{\scriptstyle F_G} \\ G^{(p)} \times_k G^{(p)} & \xrightarrow[m^{(p)}]{} & G^{(p)}. \end{array}$$

Thus F_G preserves multiplication. Functoriality applied to the unit and inverse morphisms gives their compatibility as well. □

Proposition 3.10. If G is smooth and connected, then $G^{(p)}$ is smooth and connected. Moreover, $F_G \colon G \to G^{(p)}$ is faithfully flat.

Proof. By compatibility with base change and descent, we may assume that $k = \overline{k}$. The map $f \colon k \to k$, $a \mapsto a^p$, is then an isomorphism. Hence the projection $G^{(p)} \to G$ is an isomorphism of underlying schemes, so $G^{(p)}$ is connected; it is smooth over k by base change of the smooth structure morphism of G.

The map $k \to {}_f k$, $a \mapsto a^p$, is an isomorphism of k-algebras. Hence $F_G \colon G(k) \to G^{(p)}(k)$ is bijective. All closed points of $G^{(p)}$ therefore lie in the image. The image of a homomorphism of algebraic groups is closed, so F_G is surjective. Since the target $G^{(p)}$ is smooth, hence reduced, the faithful-flatness Proposition 1.35 for surjective homomorphisms of algebraic groups applies. Thus F_G is faithfully flat. □

A Homework

A.1 Week 1

Exercise A.1. Show that the two definitions (algebraic geometry defintion and functor defintion) of algebraic groups coincide.

Proof. In the class we proved that for any algebraic group G in the algebraic geometry sense, it induces

$$\widetilde{G} : \mathrm{Alg}_k \longrightarrow \mathrm{Grp}, \qquad R \longmapsto G(R),$$

and $\widetilde{G}$ is an algebraic group in the functor sense.

It suffices to show that, for any functor $\widetilde{G} : \mathrm{Alg}_k \to \mathrm{Grp}$ such that $\widetilde{G}$, seen as $\mathrm{Alg}_k \to \mathrm{Set}$, is representable by a k-scheme G, then G is an algebraic group over k in the algebraic geometry sense.

Lemma A.2. [GW20, Corollary 4.7] Let X and Y be S-schemes. Then it is equivalent to give the following data.

1. A morphism of S-schemes from X to Y.
2. For all S-schemes T a map $f(T) : X_S(T) \to Y_S(T)$ of sets which is functorial in T, i.e., for all morphisms $u : T' \to T$ of S-schemes the following diagram is commutative:
$$\begin{array}{ccc} X_S(T) & \xrightarrow{f(T)} & Y_S(T) \\ {\scriptstyle X_S(u)}\downarrow & & \downarrow{\scriptstyle Y_S(u)} \\ X_S(T') & \xrightarrow[f(T')]{} & Y_S(T'). \end{array}$$
3. For all affine S-schemes $T = \operatorname{Spec} B$ a map $f(T) : X_S(T) \to Y_S(T)$ of sets which is functorial in B.

- **Construction of $G \times G \xrightarrow{m} G$.** By Lemma A.2, it is equivalent to give, for all $R \in \mathrm{Alg}_k$, a map
$$(G \times G)(R) \xrightarrow{m(R)} G(R)$$

and, for all k-algebra homomorphisms $R \to T$, to have the following diagram commute:

$$\begin{array}{ccc} (G \times G)(R) & \xrightarrow{m(R)} & G(R) \\ \downarrow & & \downarrow \\ (G \times G)(T) & \xrightarrow[m(T)]{} & G(T). \end{array}$$

Because it is well-known that $(G \times G)(R) = G(R) \times G(R)$ and $G(R) \cong \widetilde{G}(R)$ is a group, $m(R)$ can just be the multiplication of $\widetilde{G}(R)$. Because the vertical maps are group homomorphisms, hence it commutes.

- **Construction of $e : \operatorname{Spec} k \to G$.** It just corresponds to $G(k) \cong \widetilde{G}(k)$, namely the element $1_{\widetilde{G}(k)}$.
- **Construction of $i : G \to G$.** By Lemma A.2, it is equivalent to give, for all $R \in \mathrm{Alg}_k$, a map

$$G(R) \xrightarrow{i(R)} G(R)$$

and, for all k-algebra homomorphisms $R \to T$, to have the following diagram commute:

$$\begin{array}{ccc} G(R) & \xrightarrow{i(R)} & G(R) \\ \downarrow & & \downarrow \\ G(T) & \xrightarrow[i(T)]{} & G(T). \end{array}$$

Since $G(R) \cong \widetilde{G}(R)$ is a group, $i(R)$ can just be the inverse map of $\widetilde{G}(R)$. Because the vertical maps are group homomorphisms, hence it commutes.

- **Associativity.** To prove that

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{(m,\mathrm{id}_G)} & G \times G \\ {\scriptstyle (\mathrm{id}_G,m)}\downarrow & & \downarrow{\scriptstyle m} \\ G \times G & \xrightarrow[m]{} & G \end{array}$$

commutes, i.e.,

$$G \times G \times G \xrightarrow{(m,\mathrm{id}_G)} G \times G \xrightarrow{m} G$$

and

$$G \times G \times G \xrightarrow{(\mathrm{id}_G,m)} G \times G \xrightarrow{m} G$$

give the same morphism, by Lemma A.2 it suffices to show that for every $R \in \mathrm{Alg}_k$,

$$(G \times G \times G)(R) \xrightarrow{(m(R),\mathrm{id}_{G(R)})} (G \times G)(R) \xrightarrow{m(R)} G(R)$$

and

$$(G \times G \times G)(R) \xrightarrow{(\mathrm{id}_{G(R)},m(R))} (G \times G)(R) \xrightarrow{m(R)} G(R)$$

give the same map. It is obvious by $G(R) \cong \widetilde{G}(R)$ being a group and the construction of $m(R)$, i.e., it is just the associativity on $\widetilde{G}(R)$.

- **Existence of unit.** To show

$$\begin{array}{ccc} G \cong G \times \operatorname{Spec} k & \xrightarrow{(\mathrm{id}_G,e)} & G \times G \\ & \searrow^{\mathrm{id}_G} & \downarrow m \\ & & G \end{array}$$

commutes, it suffices to show for every $R \in \mathrm{Alg}_k$ that

$$\begin{array}{ccc} G(R) & \xrightarrow{g\mapsto(g,1)} & G(R) \times G(R) \\ & \searrow^{\mathrm{id}_{G(R)}} & \downarrow m(R) \\ & & G(R) \end{array}$$

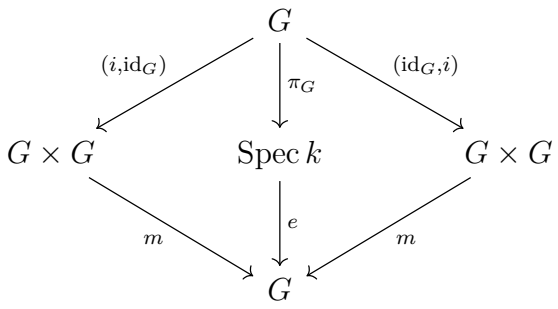
commutes, obviously.

Similarly,

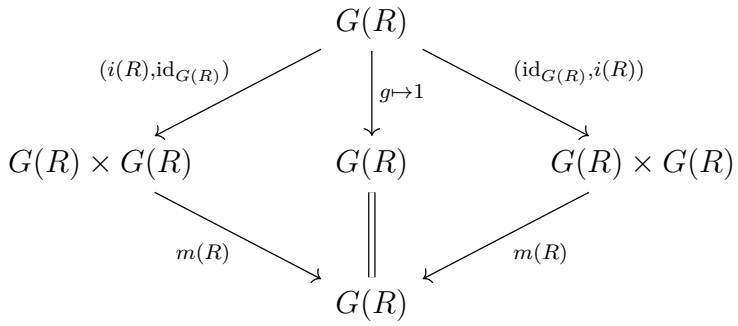
$$\begin{array}{ccc} G & \xrightarrow{(e,\mathrm{id}_G)} & G \times G \\ & \searrow^{\mathrm{id}_G} & \downarrow m \\ & & G \end{array}$$

commutes can be proved similarly.

- **Inverse property.** To show the following diagram commutes,



it suffices to show, for every $R \in \mathrm{Alg}_k$, the following diagram of maps commutes:



namely $g \mapsto (g^{-1}, g) \mapsto 1$ and $g \mapsto (g, g^{-1}) \mapsto 1$, obviously.

□

Exercise A.3. (a) Show that GL_n, B_n and U_n are linear algebraic groups.

(b) Show that U_n and B_n are both algebraic subgroups of GL_n.

(c) write down $m^\#$, $i^\#$ and $e^\#$ for GL_2.

(d) Define the functor $\mathrm{SL}_n : \mathrm{Alg}_k \to (\mathrm{gps})$ such that

$$\mathrm{SL}_n(R) = \{A \in \mathrm{GL}_n(R) \mid \det(A) = 1\}.$$

Show that it's a linear algebraic group, called special linear group.

(e) Compute the dimenson of all these algebraic groups GL_n, SL_n, B_n and U_n. Here dimension of an algebraic group is defined to be the dimension of the underlying scheme over k.

Proof of (a). **Claim.** See GL_n, B_n, U_n as functors from Alg_k to Grp as

$$\begin{aligned}\mathrm{GL}_n(R) &= \{(x_{ij}) \in M_n(R) \mid \det(x_{ij}) \in R^\times\},\\ B_n(R) &= \{(x_{ij}) \in \mathrm{GL}_n(R) \mid x_{ij} = 0 \text{ for } i > j\},\\ U_n(R) &= \{(x_{ij}) \in B_n(R) \mid x_{ii} = 1 \text{ for all } i\}.\end{aligned}$$

GL_n (B_n, U_n) is represented by affine $\mathrm{GL}_{n,k}$ $(B_{n,k}, U_{n,k})$. In fact,

$$\mathrm{GL}_{n,k} = \mathrm{Spm}\, k[X_{ij}]_{\det(X_{ij})},$$

where $\det(X_{ij})$ is a polynomial in $k[X_{ij}]$ and the subscript denotes localization,

$$B_{n,k} = \mathrm{Spm}\left(k[X_{ij}]_{\det(X_{ij})}/(X_{ij})_{i>j}\right) = \mathrm{Spm}\left((k[X_{ij}]_{i\le j})_{X_{11}X_{22}\cdots X_{nn}}\right),$$

and

$$U_{n,k} = \mathrm{Spm}\left(\frac{(k[X_{ij}]_{i\le j})_{X_{11}X_{22}\cdots X_{nn}}}{(X_{11}-1, X_{22}-1, \ldots, X_{nn}-1)}\right) \cong \mathrm{Spm}(k[X_{ij}]_{i<j}).$$

Reason: it suffices to show that

$$\mathrm{GL}_n(R) \cong \mathrm{Hom}_{\mathrm{Sch}}(\mathrm{Spm}\, R, \mathrm{GL}_{n,k}) \cong \mathrm{Hom}_{\mathrm{Alg}_k}(k[X_{ij}]_{\det(X_{ij})}, R), \tag{1}$$

$$B_n(R) \cong \mathrm{Hom}_{\mathrm{Alg}_k}\left((k[X_{ij}]_{i\le j})_{X_{11}\cdots X_{nn}}, R\right), \tag{2}$$

$$U_n(R) \cong \mathrm{Hom}_{\mathrm{Alg}_k}(k[X_{ij}]_{i<j}, R). \tag{3}$$

Notice that

$$M_n(R) \cong \mathrm{Hom}_{\mathrm{Alg}_k}(k[X_{ij}], R), \qquad (r_{ij}) \longleftrightarrow (X_{ij} \mapsto r_{ij}).$$

Thus (1) and (2) follow from the universal property of localization:

$$\begin{array}{ccc} k[X_{ij}] & \xrightarrow{X_{ij}\mapsto r_{ij}} & R \\ \downarrow & \nearrow & \\ k[X_{ij}]_{\det(X_{ij})} & & \end{array} \qquad \text{with } \det(r_{ij}) \in R^\times,$$

and (3) is direct. □

Proof of (b). By our construction in (a),

$$U_n \hookrightarrow B_n \hookrightarrow \mathrm{GL}_n.$$

Show $U_n \hookrightarrow B_n \hookrightarrow \mathrm{GL}_n$ are homomorphisms of algebraic groups in the functorial sense; i.e.,

for every $R \in \mathrm{Alg}_k$,

$$U_n(R) \hookrightarrow B_n(R) \hookrightarrow \mathrm{GL}_n(R)$$

are homomorphisms of groups (inclusion). It is natural.

But there is a detail: the inclusion $B_n(R) \to \mathrm{GL}_n(R)$ for any $R \in \mathrm{Alg}_k$, which is obviously functorial on Alg_k, just corresponds to the natural closed immersion $B_n \hookrightarrow \mathrm{GL}_n$ in Lemma A.2.

Reason: The natural closed immersion $B_n \hookrightarrow \mathrm{GL}_n$ corresponds to the natural quotient map

$$S \longrightarrow S/I, \qquad S = k[X_{ij}]_{\det(X_{ij})}, \qquad I \text{ generated by } (X_{ij})_{i>j}.$$

For every $R \in \mathrm{Alg}_k$, the induced

$$\mathrm{Mor}(\mathrm{Spm}\, R, B_n) \longrightarrow \mathrm{Mor}(\mathrm{Spm}\, R, \mathrm{GL}_n)$$

corresponds to

$$\begin{array}{ccc} S & \xrightarrow{\text{natural}} & S/I \\ & \searrow & \downarrow \\ & & R, \end{array}$$

and corresponds to $B_n(R) \to \mathrm{GL}_n(R)$ by

$$(r_{ij}) \longmapsto (r_{ij}),$$

that is,

$$\begin{pmatrix} X_{ij} \mapsto r_{ij} \\ S/I \to R \end{pmatrix} \longmapsto \begin{pmatrix} X_{ij} \mapsto r_{ij} \\ S \to R \end{pmatrix},$$

which is just the natural inclusion. □

Solution of (c).

$$m^{\#} : k[x,y,z,w]_{(xw-yz)} \longrightarrow k[x,y,z,w]_{(xw-yz)} \otimes k[x,y,z,w]_{(xw-yz)}.$$

See an element in $k[x,y,z,w]_{(xw-yz)}$ as a function on GL_2. For example, if $m^{\#}(x) = \sum_i f_i \otimes g_i$, then for

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}, \quad \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \in \mathrm{GL}_2,$$

we have

$$a_1a_2 + b_1c_2 = x\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}\right) = \sum_i f_i\left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}\right) g_i\left(\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}\right).$$

Thus

$$m^{\#}(x) = x \otimes x + y \otimes z.$$

Similarly,

$$\begin{aligned} m^{\#}(y) &= x \otimes y + y \otimes w, \\ m^{\#}(z) &= z \otimes x + w \otimes z, \\ m^{\#}(w) &= z \otimes y + w \otimes w. \end{aligned}$$

Also

$$e^{\#} : k[x,y,z,w]_{(xw-yz)} \longrightarrow k, \qquad x, w \longmapsto 1, \quad y, z \longmapsto 0,$$

corresponding to $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and

$$i^{\#} : k[x,y,z,w]_{(xw-yz)} \longrightarrow k[x,y,z,w]_{(xw-yz)},$$

with

$$\begin{aligned} x &\longmapsto (xw-yz)^{-1}w, \\ y &\longmapsto -(xw-yz)^{-1}y, \\ z &\longmapsto -(xw-yz)^{-1}z, \\ w &\longmapsto (xw-yz)^{-1}x, \end{aligned} \qquad \text{corresponding to } A \longmapsto A^{-1}.$$

□

Proof of (d). $\mathrm{SL}_n(R)$ are groups for any $R \in \mathrm{Alg}_k$. It suffices to show SL_n, as a functor from Alg_k to Set, is represented by

$$\mathrm{SL}_{n,k} := \mathrm{Spm}\left(\frac{k[X_{ij}]_{\det(X_{ij})}}{(\det(X_{ij})-1)}\right) = \mathrm{Spm}\left(\frac{k[X_{ij}]}{(\det(X_{ij})-1)}\right).$$

It suffices to show that, for any $R \in \mathrm{Alg}_k$,

$$\mathrm{SL}_n(R) \cong \mathrm{Hom}_{\mathrm{Sch}}(\mathrm{Spm}\, R, \mathrm{SL}_n) \cong \mathrm{Hom}_{\mathrm{Alg}_k}\left(\frac{k[X_{ij}]}{(\det(X_{ij})-1)}, R\right),$$

and this is satisfied by

$$(r_{ij}) \longleftrightarrow (X_{ij} \mapsto r_{ij}).$$

□

Solution of (e). GL_n is an open subscheme of $\operatorname{Spm} k[X_{ij}]$, hence

$$\dim \mathrm{GL}_n = n^2.$$

SL_n is a hypersurface of $\operatorname{Spm} k[X_{ij}]$, hence

$$\dim \mathrm{SL}_n = n^2 - 1.$$

B_n is cut out of GL_n by $\frac{n(n-1)}{2}$ functions X_{ij}, $i > j$. Hence

$$\dim B_n = n^2 - \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

Finally,

$$U_n \cong \operatorname{Spm} k[X_{ij}]_{i<j}, \qquad \dim U_n = \frac{n(n-1)}{2}.$$

□

Exercise A.4. For $C \in \mathrm{GL}_n(k)$, consider the functor $G_C : \mathrm{Alg}_k \to (\mathrm{gps})$ with

$$G_C(R) = \{A \in \mathrm{GL}_n(R) \mid A^t C A = C\}.$$

(a). Show that G_C is a linear algebraic group.

(b). When $n = 2m$ is even and $C = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}$, then G_C is called the symplectic group denoted by Sp_n. Compute its dimension.

(c). For $\mathrm{char}(k) \neq 2$, when $C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_m \\ 0 & I_m & 0 \end{pmatrix}$ (resp. $C = \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}$) if $n = 2m+1$ odd (resp. $n = 2m$ even), then G_C is called the orthogonal group denoted by O_n. Compute its dimension.

Proof of (a). $G_C(R)$ is a group for all $R \in \mathrm{Alg}_k$. Write

$$C = (c_{ij}), \qquad c_{ij} \in k.$$

Then

$$(X_{ij})^t (c_{ij})(X_{ij}) - (c_{ij}) =: \big(f_{ij}(X_{11}, \ldots, X_{nn})\big), \qquad f_{ij} \in k[X_{ij}].$$

Claim. $G_C(-)$ is represented by

$$\operatorname{Spm} \frac{k[X_{ij}]}{(f_{ij})},$$

which is affine.

According to the definition, at first it should be

$$\operatorname{Spm}\left(k[X_{ij}]_{\det(X_{ij})}/(f_{ij})\right),$$

but $\det(X_{ij})$ can be removed because $A^tCA = C$ and $C \in \mathrm{GL}_n$ imply $A \in \mathrm{GL}_n$.

Reason: It suffices to show, for any $R \in \mathrm{Alg}_k$,

$$G_C(R) \cong \operatorname{Hom}_{\mathrm{Sch}}\left(\operatorname{Spm} R, \operatorname{Spm} \frac{k[X_{ij}]}{(f_{ij})}\right) \cong \operatorname{Hom}_{\mathrm{Alg}_k}\left(\frac{k[X_{ij}]}{(f_{ij})}, R\right),$$

and it is satisfied by

$$(r_{ij}) \longleftrightarrow (X_{ij} \mapsto r_{ij}).$$

□

Solution of (b). Notice that, because C is anti-symmetric,

$$f_{ij} = -f_{ji}.$$

Therefore $(f_{ij}) \subseteq k[X_{ij}]$ can be generated by at most $\frac{n(n-1)}{2}$ polynomials. Hence

$$\dim \mathrm{Sp}_n \geq n^2 - \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

We want to compute the corank of the Jacobian to get the upper bound of $\dim \mathrm{Sp}_n$. By [Vak25, 13.1.M, 13.1.N], for every $p \in \mathrm{Sp}_n$,

$$\begin{aligned}\dim_{\mathrm{Sp}_n} p &\leq \operatorname{corank}\left(\left(\frac{\partial f_{ij}}{\partial X_{kl}}(p)\right)_{i,j,k,l\leq n}\right) \\ &= \dim_{\kappa(p)}\left\{B = (b_{kl}) \in M_{n\times n}(\kappa(p)) \;\middle|\; \forall i,j,\ \sum_{k,l} \frac{\partial f_{ij}}{\partial X_{kl}}(p)\cdot b_{kl} = 0\right\}.\end{aligned}$$

By

$$f_{ij} = \sum_{r,s} X_{ri} c_{rs} X_{sj} - c_{ij},$$

we have, for every k, l,

$$\frac{\partial f_{ij}}{\partial X_{kl}} = \delta_{il} \sum_s c_{ks} X_{sj} + \delta_{jl} \sum_r X_{ri} c_{rk}.$$

Thus, for every i, j,

$$\begin{aligned} \sum_{k,l} \frac{\partial f_{ij}}{\partial X_{kl}}(p) \cdot b_{kl} &= \sum_{k,s} b_{ki} c_{ks} X_{sj}(p) + \sum_{r,k} X_{ri}(p) c_{rk} b_{kj} \\ &= (B^t C A(p) + A(p)^t C B)_{ij}. \end{aligned}$$

Therefore

$$\begin{aligned} \dim_{\mathrm{Sp}_n} p &\le \dim_{\kappa(p)} \{B \in M_{n\times n}(\kappa(p)) \mid B^t C A(p) + A(p)^t C B = 0\} \\ &= \dim_{\kappa(p)} \{B \in M_{n\times n}(\kappa(p)) \mid B^t C A(p) = (B^t C A(p))^t\} \qquad \text{by } C = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix} \\ &= \dim_{\kappa(p)} \{Y \in M_{n\times n}(\kappa(p)) \mid Y = Y^t\} \qquad \text{by } C, A(p) \in \mathrm{GL}_n(\kappa(p)) \\ &= \frac{n(n+1)}{2}. \end{aligned}$$

Hence

$$\dim \mathrm{Sp}_n = \frac{n(n+1)}{2}.$$

□

Solution of (c). Because C is symmetric,

$$f_{ij} = f_{ji}.$$

Therefore $(f_{ij}) \subseteq k[X_{ij}]$ can be generated by at most $\frac{n(n+1)}{2}$ polynomials. Hence

$$\dim \mathrm{O}_n \ge n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

By a similar discussion as in (b), for every $p \in \mathrm{O}_n$,

$$\begin{aligned} \dim_{\mathrm{O}_n} p &\le \dim_{\kappa(p)} \{B \in M_{n\times n}(\kappa(p)) \mid B^t C A(p) + A(p)^t C B = 0\} \\ &= \dim_{\kappa(p)} \{B \in M_{n\times n}(\kappa(p)) \mid B^t C A(p) = -(B^t C A(p))^t\} \qquad \text{by } C = C^t \text{ and } \operatorname{char} k \neq 2 \\ &= \dim_{\kappa(p)} \{Y \in M_{n\times n}(\kappa(p)) \mid Y = -Y^t\} \qquad \text{by } C, A(p) \in \mathrm{GL}_n(\kappa(p)) \\ &= \frac{n(n-1)}{2}. \end{aligned}$$

Hence

$$\dim \mathrm{O}_n = \frac{n(n-1)}{2}.$$

□

Exercise A.5. Show that the sequence of homomomorphism of algebraic groups

$$e \to \mathrm{SL}_n \to \mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m \to e$$

is exact, here the first map is the natural inclusion and the second one is the determinant map.

Proof.

- $\mathrm{SL}_n \cong \mathrm{Ker}(\det)$. For every $R \in \mathrm{Alg}_k$,

  $$K(R) := \ker\big(\mathrm{GL}_n(R) \xrightarrow{\det} \mathbb{G}_m(R) = R^\times\big)$$

  is a group. By definition, $K(R)$ is represented by $\mathrm{Ker}(\mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m)$. Also, $K(R) = \mathrm{SL}_n(R)$ is represented by SL_n.

- $\mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m$ is faithfully flat, i.e. flat and surjective by [Vak25, The paragraph after 24.5.B].

  Surjectivity. It suffices to find a section $s : \mathbb{G}_m \to \mathrm{GL}_n$. It suffices to find a k-algebra homomorphism

  $$s^\# : k[X_{ij}]_{\det(X_{ij})} \longrightarrow k[t, t^{-1}]$$

  such that, with

  $$\overset{\#}{\det} : k[t, t^{-1}] \longrightarrow k[X_{ij}]_{\det(X_{ij})}, \qquad t \longmapsto \det(X_{ij}),$$

  we have

  $$s^\# \circ \overset{\#}{\det} = \mathrm{id}_{k[t,t^{-1}]}.$$

  Consider $s^\#$ given by

  $$X_{11} \longmapsto t, \qquad X_{ii} \longmapsto 1 \text{ for } i > 1, \qquad X_{ij} \longmapsto 0 \text{ for } i \neq j.$$

  So

  $$\mathbb{G}_m \xrightarrow{s} \mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m, \qquad \mathbb{G}_m \xrightarrow{\mathrm{id}} \mathbb{G}_m,$$

  and det is surjective.

Flatness. With

$$k[t,t^{-1}] \xrightarrow{\det^{\#}} k[X_{ij}]_{\det(X_{ij})}$$

injective by the above choice of the section and the fact that $k[X_{ij}]_{\det(X_{ij})}$ is integral, $k[X_{ij}]_{\det(X_{ij})}$ is torsion-free as a $k[t,t^{-1}]$-module. The ring $k[t,t^{-1}]$ is a localization of a PID, hence is a PID. A torsion-free algebra over a PID is flat over it.

□

A.2 Week 2

Exercise A.6. The natural isomorphism of set valued functors

$$\mathrm{SL}_n(R) \times \mathbb{G}_m(R) \longrightarrow \mathrm{GL}_n(R), \qquad (A,r) \longmapsto A \cdot \mathrm{diag}(r,1,\ldots,1)$$

defines an isomorphism of k-algebras

$$\mathcal{O}(\mathrm{GL}_n) \simeq \mathcal{O}(\mathrm{SL}_n) \otimes \mathcal{O}(\mathbb{G}_m).$$

(a) Show that SL_n is connected.

(b) Please describe the map

$$\mathcal{O}(\mathrm{SL}_n) \longrightarrow \mathcal{O}(\mathrm{GL}_n)$$

from the previous isomorphism.

Proof. **(a)** By the given isomorphism and Yoneda,

$$\mathrm{SL}_n \times \mathbb{G}_m \simeq \mathrm{GL}_n.$$

Also

$$\mathrm{GL}_n \simeq \mathrm{Spm}\, k[X_{ij}]_\Delta, \qquad \Delta = \det(X_{ij}).$$

Since $k[X_{ij}]_\Delta$ is an integral domain, GL_n is irreducible, hence connected. Consider

$$\mathrm{GL}_n \xrightarrow{\sim} \mathrm{SL}_n \times \mathbb{G}_m \xrightarrow{\mathrm{pr}_1} \mathrm{SL}_n.$$

It is surjective. The image of a connected space is connected, so SL_n is connected.

(b) The map in question is induced by

$$\mathrm{GL}_n \xrightarrow{\sim} \mathrm{SL}_n \times \mathbb{G}_m \xrightarrow{\mathrm{pr}_1} \mathrm{SL}_n.$$

On coordinate rings it is

$$\mathcal{O}(\mathrm{SL}_n) \longrightarrow \mathcal{O}(\mathrm{SL}_n) \otimes \mathcal{O}(\mathbb{G}_m) \xrightarrow{\sim} \mathcal{O}(\mathrm{GL}_n), \qquad f \longmapsto f \otimes 1.$$

Write

$$\mathcal{O}(\mathrm{SL}_n) = \frac{k[X_{ij}]}{(\Delta - 1)}, \qquad \mathcal{O}(\mathbb{G}_m) = k[T, T^{-1}], \qquad \mathcal{O}(\mathrm{GL}_n) = k[X_{ij}]_\Delta,$$

and let $\overline{X}_{ij}$ denote the image of X_{ij} in $\mathcal{O}(\mathrm{SL}_n)$. Under the given functorial isomorphism,

$$X_{i1} \longmapsto \overline{X}_{i1} \otimes T, \qquad X_{ij} \longmapsto \overline{X}_{ij} \otimes 1 \quad (j \neq 1), \qquad \Delta \longmapsto 1 \otimes T.$$

Therefore

$$\boxed{\overline{X}_{i1} \longmapsto \frac{X_{i1}}{\Delta}, \qquad \overline{X}_{ij} \longmapsto X_{ij} \quad (j \neq 1).}$$

□

Exercise A.7. Let $\mathrm{char}(k) \neq 2$. Let $O' = G_C$ with $C = I_n$ (notation from Week 1 Ex. 3).

(a) Show that

$$O_{n,\overline{k}} \simeq O'_{n,\overline{k}}$$

as algebraic groups over $\overline{k}$.

(b) Show that the determinant map

$$\det : O_n \longrightarrow \mu_2$$

is a homomorphism of algebraic groups, where

$$\mu_2 = \mathrm{Ker}(\mathbb{G}_m \longrightarrow \mathbb{G}_m,\ x \longmapsto x^2).$$

Define

$$\mathrm{SO}_n := \mathrm{Ker}(\det : O_n \to \mu_2).$$

And similarly define

$$\mathrm{SO}'_n := \mathrm{Ker}(\det : O'_n \to \mu_2).$$

(c) For each k-algebra R, let $V(R)$ denote the set of skew-symmetric matrices, i.e.

the matrices $A \in M_n(R)$ such that $A^t = -A$. Show that the functor

$$R \longmapsto V(R)$$

is represented by a finitely generated k-algebra C, and that C is an integral domain.

(d) Show that

$$A \longmapsto (I_n - A)(I_n + A)^{-1}$$

defines a bijection from a nonempty open subset of $\mathrm{SO}'_n(\overline{k})$ onto an open subset of $V(\overline{k})$, with partial inverse

$$B \longmapsto (I_n - B)(I_n + B)^{-1}.$$

(e) Deduce that $\mathrm{SO}'_{n,\overline{k}}$ is connected.

(f) Deduce that SO_n is connected.

Proof. Let J be the symmetric matrix defining O_n in Week 1, namely

$$J = \begin{cases} \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}, & n = 2m, \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_m \\ 0 & I_m & 0 \end{pmatrix}, & n = 2m+1. \end{cases}$$

(a) Choose $\alpha, \beta \in \overline{k}$ such that

$$\alpha^2 = 2, \qquad \beta^2 = -1.$$

For the 2×2 hyperbolic block we have

$$\begin{pmatrix} \frac{1}{\alpha} & \frac{1}{\alpha} \\ \frac{\beta}{\alpha} & -\frac{\beta}{\alpha} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\alpha} & \frac{\beta}{\alpha} \\ \frac{1}{\alpha} & -\frac{\beta}{\alpha} \end{pmatrix} = I_2.$$

Applying this to the hyperbolic blocks of J gives $Q \in \mathrm{GL}_n(\overline{k})$ such that

$$Q^t J Q = I_n.$$

Hence, for every $R \in \mathrm{Alg}_{\overline{k}}$,

$$O_{n,\overline{k}}(R) = \{A \in \mathrm{GL}_n(R) \mid A^t J A = J\}$$

is naturally identified with

$$\{B \in \mathrm{GL}_n(R) \mid B^t I_n B = I_n\} = O'_{n,\overline{k}}(R)$$

by

$$A \longmapsto Q^{-1} A Q.$$

Indeed, $Q^t J Q = I_n$ implies $J = Q^{-t} Q^{-1}$, and therefore

$$(Q^{-1}AQ)^t (Q^{-1}AQ) = Q^t A^t J A Q = Q^t J Q = I_n.$$

The inverse is $B \mapsto QBQ^{-1}$. This is functorial in R, so by Yoneda

$$O_{n,\overline{k}} \simeq O'_{n,\overline{k}}.$$

(b) It is enough to show that for every $R \in \mathrm{Alg}_k$,

$$\det(R) : O_n(R) \longrightarrow \mu_2(R)$$

is a group homomorphism. We have

$$\mu_2(R) = \mathrm{Ker}\Big(\mathbb{G}_m(R) \xrightarrow{r \mapsto r^2} \mathbb{G}_m(R)\Big) = \{r \in R^\times \mid r^2 = 1\}.$$

For $A \in O_n(R)$, from

$$A^t J A = J$$

we get

$$\det(A)^2 \det(J) = \det(J).$$

Since $\det(J) \in k^\times$, it follows that

$$\det(A)^2 = 1.$$

Thus the image of $O_n(R)$ under determinant is contained in $\mu_2(R)$. Moreover,

$$O_n(R) \xrightarrow{\det} \mathbb{G}_m(R)$$

is a group homomorphism. Hence

$$O_n(R) \longrightarrow \mu_2(R)$$

is a group homomorphism for every R, and therefore

$$\det : O_n \longrightarrow \mu_2$$

is a homomorphism of algebraic groups.

(c) We claim that V is represented by

$$\operatorname{Spm} \frac{k[X_{ij} \mid 1 \le i,j \le n]}{(X_{ij} + X_{ji} \mid 1 \le i,j \le n)}.$$

Indeed, for every $R \in \mathrm{Alg}_k$,

$$\begin{aligned}\operatorname{Hom}_{k\text{-alg}}\left(\frac{k[X_{ij}]}{(X_{ij}+X_{ji})}, R\right) &\simeq \{f \in \operatorname{Hom}_{k\text{-alg}}(k[X_{ij}], R) \mid f(X_{ij}) + f(X_{ji}) = 0 \ \forall i,j\} \\ &\simeq \{(r_{ij}) \in M_n(R) \mid r_{ij} + r_{ji} = 0 \ \forall i,j\} \\ &= V(R).\end{aligned}$$

Thus the claim follows by Yoneda. Since $\mathrm{char}(k) \neq 2$, the relations imply

$$X_{ii} = 0, \qquad X_{ji} = -X_{ij},$$

and hence

$$C := \frac{k[X_{ij}]}{(X_{ij}+X_{ji})} \simeq k[X_{ij} \mid i < j].$$

Therefore C is finitely generated and is an integral domain.

(d) Consider, as classical varieties,

$$U = \left\{A \in \mathrm{SO}'_n(\overline{k}) \mid \det(I_n + A) \neq 0\right\}$$

and

$$W = \left\{B \in V(\overline{k}) \mid \det(I_n + B) \neq 0\right\}.$$

These are open, and they are nonempty since $I_n \in U$ and $0 \in W$. We claim that

$$U \xrightarrow{\sim} W, \qquad A \longmapsto (I_n - A)(I_n + A)^{-1},$$

with inverse

$$B \longmapsto (I_n - B)(I_n + B)^{-1}.$$

First, it is enough to check the bijection set-theoretically. Indeed, the entries of $(I_n - Y)(I_n + Y)^{-1}$ belong to the localization in which $\det(I_n + Y)$ is inverted, so they define the corresponding homomorphism of localized $\overline{k}$-algebras. The other direction is defined similarly. Thus, once the two formulas are checked to be inverse on points, they give inverse morphisms of the two open subschemes.

Let $A \in U$ and put

$$B = (I_n - A)(I_n + A)^{-1}.$$

Since $A^t A = I_n$, we have $A^t = A^{-1}$, and

$$\begin{aligned} B^t &= (I_n + A^t)^{-1}(I_n - A^t) \\ &= (I_n + A^{-1})^{-1}(I_n - A^{-1}) \\ &= (A + I_n)^{-1}(A - I_n) \\ &= (A - I_n)(A + I_n)^{-1} \\ &= -B. \end{aligned}$$

Hence $B \in V(\overline{k})$. Moreover,

$$I_n + B = 2(I_n + A)^{-1},$$

so $\det(I_n + B) \neq 0$, and hence $B \in W$.

Conversely, let $B \in W$ and put

$$A = (I_n - B)(I_n + B)^{-1}.$$

Since $B^t = -B$,

$$\begin{aligned} A^t &= (I_n + B^t)^{-1}(I_n - B^t) \\ &= (I_n - B)^{-1}(I_n + B) \\ &= A^{-1}. \end{aligned}$$

Also

$$\det(A) = \frac{\det(I_n - B)}{\det(I_n + B)} = \frac{\det((I_n - B)^t)}{\det(I_n + B)} = \frac{\det(I_n + B)}{\det(I_n + B)} = 1.$$

Thus $A \in \mathrm{SO}'_n(\overline{k})$. Moreover,

$$I_n + A = 2(I_n + B)^{-1},$$

so $A \in U$.

Finally, if $B = (I_n - A)(I_n + A)^{-1}$, then

$$I_n - B = 2A(I_n + A)^{-1}, \qquad I_n + B = 2(I_n + A)^{-1},$$

and hence

$$(I_n - B)(I_n + B)^{-1} = A.$$

The other composition is the same. Therefore $U \simeq W$.

(e) We use the Cartan–Dieudonné theorem: for an n-dimensional nondegenerate symmetric bilinear space over a field of characteristic not 2, every element of the orthogonal group is a composition of at most n reflections.

For a nonisotropic vector v, write

$$S_v(x) = x - \frac{2\langle x, v\rangle}{\langle v, v\rangle} v.$$

Its eigenvalues are $-1, 1, \ldots, 1$, so $\det(S_v) = -1$.

Let U be the open subset from part **(d)**. We claim that for any nonisotropic vectors u, v,

$$S_u S_v \in U(\overline{k})^2.$$

Because $\overline{k}$ is infinite, we can choose a nonisotropic vector w such that

$$\langle u, w\rangle \neq 0, \qquad \langle v, w\rangle \neq 0.$$

Indeed, we choose w outside the union of the three proper closed subsets cut out by

$$\langle w, w\rangle, \qquad \langle u, w\rangle, \qquad \langle v, w\rangle.$$

We show that $S_u S_w \in U(\overline{k})$. Suppose, to the contrary, that -1 is an eigenvalue of $S_u S_w$. Choose $x \neq 0$ with

$$S_u S_w x = -x.$$

Since $S_u^2 = 1$, this gives

$$S_w x = -S_u x.$$

Using the definitions of the two reflections,

$$x - \frac{2\langle x, w\rangle}{\langle w, w\rangle} w = -x + \frac{2\langle x, u\rangle}{\langle u, u\rangle} u,$$

so

$$x = \frac{\langle x, u\rangle}{\langle u, u\rangle}u + \frac{\langle x, w\rangle}{\langle w, w\rangle}w.$$

Pairing this equality first with u and then with w gives

$$\frac{\langle x, w\rangle}{\langle w, w\rangle}\langle u, w\rangle = 0, \qquad \frac{\langle x, u\rangle}{\langle u, u\rangle}\langle u, w\rangle = 0.$$

Since $x \neq 0$, the two coefficients on the left cannot both be zero. Hence $\langle u, w\rangle = 0$, a contradiction. Therefore -1 is not an eigenvalue of $S_u S_w$, so

$$\det(I_n + S_u S_w) \neq 0.$$

Since $\det(S_u S_w) = 1$, we have $S_u S_w \in U(\overline{k})$. Similarly, $S_w S_v \in U(\overline{k})$. Hence

$$S_u S_v = (S_u S_w)(S_w S_v) \in U(\overline{k})^2.$$

Now let $A \in \mathrm{SO}'_n(\overline{k})$. By Cartan–Dieudonné,

$$A = S_{v_1} \cdots S_{v_\ell}.$$

Since

$$1 = \det(A) = (-1)^\ell,$$

the integer ℓ is even. Pairing the reflections and applying the claim shows that

$$A \in \langle U(\overline{k})\rangle.$$

Therefore

$$\mathrm{SO}'_n(\overline{k}) = \langle U(\overline{k})\rangle.$$

By part **(d)**, U is isomorphic to the nonempty open subset W of V. By part **(c)**, V is irreducible. Hence U is irreducible, and therefore connected. Since $I_n \in U$, we have

$$U \subseteq (\mathrm{SO}'_{n,\overline{k}})^\circ.$$

The identity component is a subgroup, so it contains the subgroup generated by $U(\overline{k})$. As this subgroup is all of $\mathrm{SO}'_n(\overline{k})$, every $\overline{k}$-point lies in the identity component. Thus

$$\mathrm{SO}'_{n,\overline{k}} = (\mathrm{SO}'_{n,\overline{k}})^\circ,$$

and $\mathrm{SO}'_{n,\overline{k}}$ is connected.

(f) From part **(a)**, for every $R \in \mathrm{Alg}_{\overline{k}}$,

$$O_{n,\overline{k}}(R) \simeq O'_{n,\overline{k}}(R).$$

This isomorphism is given by conjugation, hence preserves determinant. Therefore it restricts functorially to

$$\mathrm{SO}_{n,\overline{k}}(R) \simeq \mathrm{SO}'_{n,\overline{k}}(R),$$

and by Yoneda

$$\mathrm{SO}_{n,\overline{k}} \simeq \mathrm{SO}'_{n,\overline{k}}.$$

By part **(e)**, $\mathrm{SO}'_{n,\overline{k}}$ is connected. Hence SO_n is geometrically connected, and in particular connected. □

Exercise A.8. Let k be algebraically closed.

(a) Let G be a closed subgroup of GL_n over k. Then $\mathbb{A}^n$ has a natural action of G. Determine the orbits and closed orbits for

$$G = \mathrm{GL}_n, T_n, \mathrm{SL}_n.$$

(b) The natural action of $G = \mathrm{GL}_2$ on $\mathbb{A}^2$ induces an action of G on $\mathbb{P}^1$ (as $\mathbb{P}^1$ is the quotient of $\mathbb{A}^2 \setminus \{(0,0)\}$). Please determine the orbits for the action of G on $\mathbb{P}^1$ and the diagonal action of G on $\mathbb{P}^1 \times \mathbb{P}^1$. Please also indicate the closed orbits.

Proof. **(a)** For $G = \mathrm{GL}_n$, the orbits are

$$\mathbb{A}^n \setminus \{0\}, \qquad \{0\}.$$

Indeed, GL_n acts transitively on the nonzero vectors. The only closed orbit is $\{0\}$.

For $G = T_n$, the orbits are indexed by subsets

$$S \subseteq \{1, \ldots, n\}.$$

For such an S, the corresponding orbit is

$$\mathcal{O}_S = \{(x_1, \ldots, x_n) \in \mathbb{A}^n \mid x_i \neq 0 \iff i \in S\}.$$

Equivalently,

$$\mathcal{O}_S \simeq \mathrm{Spm}\left(k[X_i \mid i \in S]_{\prod_{i \in S} X_i}\right) \subseteq \mathrm{Spm}\, k[X_i \mid i \in S] \subseteq \mathbb{A}^n.$$

Its closure is the corresponding coordinate subspace, so the only closed orbit is $\{0\}$.

For $G = \mathrm{SL}_n$, when $n = 1$, the group SL_1 is trivial. Hence every point is an orbit, and every orbit is closed. When $n \geq 2$, SL_n is transitive on $\mathbb{A}^n \setminus \{0\}$: extend a nonzero vector to a basis and rescale one of the remaining basis vectors so that the resulting matrix has determinant 1. Thus the orbits are

$$\mathbb{A}^n \setminus \{0\}, \qquad \{0\},$$

and the only closed orbit is $\{0\}$.

(b) By the discussion above, the orbit of $G = \mathrm{GL}_2$ on $\mathbb{P}^1$ is all of

$$\mathbb{P}^1.$$

Hence it is closed.

For the diagonal action of G on $\mathbb{P}^1 \times \mathbb{P}^1$, there are two orbits:

$$\Delta, \qquad (\mathbb{P}^1 \times \mathbb{P}^1) \setminus \Delta,$$

where

$$\Delta = \mathrm{im}\Big(\mathbb{P}^1 \xrightarrow{\text{diag}} \mathbb{P}^1 \times \mathbb{P}^1\Big).$$

Indeed, Δ is one orbit by transitivity on $\mathbb{P}^1$. If $x \neq y$, nonzero representatives of the two distinct lines form a basis of k^2, so every ordered pair of distinct points belongs to the same orbit. Since $\mathbb{P}^1$ is complete, the diagonal morphism is proper and its image Δ is closed. Therefore Δ is the only closed orbit. □

A.3 Week 3

Exercise A.9. Let $V = k^2$ be the natural representation of $G = \mathrm{GL}_2$.

(a) Let $V^\vee$ be the dual representation of V. Determine the $\mathcal{O}(G)$-comodule of $V^\vee$.

(b) Determine the $\mathcal{O}(G)$-comodule of the representation $V \otimes V$.

Proof. (a) Let $\{e_i\}_{i=1}^2$ be the standard basis of V, and let $\{e_i^\vee\}_{i=1}^2$ be its dual basis, so $e_i^\vee(e_j) = \delta_{ij}$. Write

$$\rho(e_j^\vee) = \sum_{i=1}^{2} e_i^\vee \otimes f_{ij}, \qquad f_{ij} \in \mathcal{O}(G).$$

This corresponds to the dual representation $G \longrightarrow \mathrm{GL}_{V^\vee}$. For a k-algebra R and $g \in G(R)$,

its action on $V^\vee \otimes_k R$ is given by

$$(ge_j^\vee)(w) = e_j^\vee(g^{-1}w).$$

Consequently, evaluating the coaction at g and then at e_k gives

$$f_{kj}(g) = (ge_j^\vee)(e_k) = e_j^\vee(g^{-1}e_k) = (g^{-1})_{jk}.$$

Put $\Delta = \det(T_{ij}) = T_{11}T_{22} - T_{12}T_{21}$. For $n = 2$,

$$g^{-1} = \frac{1}{\Delta(g)} \begin{pmatrix} T_{22}(g) & -T_{12}(g) \\ -T_{21}(g) & T_{11}(g) \end{pmatrix}.$$

Thus

$$f_{11} = \frac{T_{22}}{\Delta}, \qquad f_{12} = -\frac{T_{21}}{\Delta}, \qquad f_{21} = -\frac{T_{12}}{\Delta}, \qquad f_{22} = \frac{T_{11}}{\Delta},$$

and the requested coaction is

$$\rho(e_1^\vee) = e_1^\vee \otimes \frac{T_{22}}{\Delta} - e_2^\vee \otimes \frac{T_{12}}{\Delta},$$
$$\rho(e_2^\vee) = -e_1^\vee \otimes \frac{T_{21}}{\Delta} + e_2^\vee \otimes \frac{T_{11}}{\Delta}.$$

(b) Write

$$\rho(e_i \otimes e_j) = \sum_{r,s=1}^{2} (e_r \otimes e_s) \otimes f_{ij}^{rs}.$$

For every k-algebra R and $g \in G(R)$, evaluation gives

$$\sum_{r,s} (e_r \otimes e_s) f_{ij}^{rs}(g) = g(e_i \otimes e_j) = ge_i \otimes ge_j = \sum_{r,s} (e_r \otimes e_s) g_{ri} g_{sj}.$$

Therefore $f_{ij}^{rs} = T_{ri}T_{sj}$, and hence

$$\boxed{\rho(e_i \otimes e_j) = \sum_{r,s=1}^{2} (e_r \otimes e_s) \otimes T_{ri}T_{sj}.}$$

□

Exercise A.10. Let $G = \mathrm{GL}_2$. Let $V = \mathcal{O}(G)$ be the regular representation of G. Let $f \in V$ such that $f(g) = a_{11}a_{12}$ for any $g = (a_{ij})_{1 \le i,j \le 2}$. Find a finite dimensional

subrepresentation W of V such that $f \in W$.

Proof. For the regular representation the coaction is the comultiplication

$$m^{\#} : \mathcal{O}(G) \longrightarrow \mathcal{O}(G) \otimes \mathcal{O}(G), \qquad T_{ij} \longmapsto \sum_{r=1}^{2} T_{ir} \otimes T_{rj}.$$

Here $f = T_{11}T_{12}$, so

$$\begin{aligned} m^{\#}(f) &= (T_{11} \otimes T_{11} + T_{12} \otimes T_{21})(T_{11} \otimes T_{12} + T_{12} \otimes T_{22}) \\ &= T_{11}^2 \otimes T_{11}T_{12} + T_{11}T_{12} \otimes (T_{21}T_{12} + T_{11}T_{22}) + T_{12}^2 \otimes T_{21}T_{22}. \end{aligned}$$

$$\boxed{W = \operatorname{span}_k\{T_{11}^2, T_{11}T_{12}, T_{12}^2\}.}$$

Clearly $f \in W$. Moreover,

$$\begin{aligned} m^{\#}(T_{11}^2) &= T_{11}^2 \otimes T_{11}^2 + 2T_{11}T_{12} \otimes T_{11}T_{21} + T_{12}^2 \otimes T_{21}^2, \\ m^{\#}(T_{12}^2) &= T_{11}^2 \otimes T_{12}^2 + 2T_{11}T_{12} \otimes T_{12}T_{22} + T_{12}^2 \otimes T_{22}^2. \end{aligned}$$

Together with the displayed formula for $m^{\#}(f)$, these identities show that $m^{\#}(W) \subseteq W \otimes \mathcal{O}(G)$. Thus W is a finite dimensional subrepresentation containing f. □

Exercise A.11. Let G be a connected algebraic group over a field k of characteristic zero, and let H be an algebraic subgroup of G. Show that H is normal in G if and only if, for every representation (V, r) of G and character χ of H, the eigenspace V_χ is stable under G.

Proof. Suppose first that H is normal in G. We will prove, in comodule language, that

$$\rho_r(V_\chi) \subseteq V_\chi \otimes_k \mathcal{O}(G). \tag{1}$$

It is first enough to prove that every character of H is invariant under conjugation by G. Consider the conjugation morphism

$$c : H \times G \longrightarrow H, \qquad (h, g) \longmapsto g^{-1}hg,$$

and put $a = \chi^{\#}(T) \in \mathcal{O}(H)$. Its comorphism $c^{\#} : \mathcal{O}(H) \to \mathcal{O}(H) \otimes \mathcal{O}(G)$ is the $\mathcal{O}(G)$-comodule structure on $\mathcal{O}(H)$. As in the finite dimensional subcomodule construction, choose

a finite dimensional subcomodule U containing a, with a k-basis $u_1, \dots, u_N$, and write

$$c^{\#}(a) = \sum_{i=1}^{N} u_i \otimes b_i, \qquad b_i \in \mathcal{O}(G).$$

The coefficients b_i define a morphism $\varphi : G_{\bar{k}} \to U_{\bar{k}} \simeq \mathbb{A}^N_{\bar{k}}$. For $g \in G(\bar{k})$, the corresponding specialization is

$$c^{\#}_g(a) = \sum_{i=1}^{N} u_i \, b_i(g) \quad \text{in } \mathcal{O}(H)_{\bar{k}}$$

is group-like: it is the element corresponding to the character $h \mapsto \chi(g^{-1}hg)$. Distinct group-like elements are linearly independent. Therefore only finitely many group-like elements can belong to $U_{\bar{k}}$. Thus $\varphi\big(G(\bar{k})\big)$ is finite. Since G is connected, it is geometrically irreducible; in characteristic zero $G_{\bar{k}}$ is reduced, so $G(\bar{k})$ is Zariski dense in $G_{\bar{k}}$. It follows that φ is constant. At the identity it has value a; therefore $c^{\#}(a) = a \otimes 1$. Hence the comorphisms of $\chi \circ c$ and $\chi \circ \mathrm{pr}_H$ agree on T, and so

$$\chi \circ c = \chi \circ \mathrm{pr}_H, \qquad \text{i.e.} \qquad \chi_R(g^{-1}hg) = \chi_R(h) \quad (R \in \mathrm{Alg}_k).$$

Now let $v \in V_\chi$, $R \in \mathrm{Alg}_k$, $g \in G(R)$, and $h \in H(R)$. Normality gives $g^{-1}hg \in H(R)$, and hence

$$h(gv) = g\big((g^{-1}hg)v\big) = gv\,\chi_R(g^{-1}hg) = gv\,\chi_R(h). \tag{2}$$

Write $\rho_r(v) = \sum_i v_i \otimes f_i$ with the $f_i \in \mathcal{O}(G)$ linearly independent. Applying (2) to the universal points of H and G[10]gives

$$\sum_i \rho_{r|_H}(v_i) \otimes f_i = \sum_i (v_i \otimes \chi^{\#}(T)) \otimes f_i \quad \text{in } V \otimes \mathcal{O}(H) \otimes \mathcal{O}(G).$$

Since the f_i are linearly independent, $\rho_{r|_H}(v_i) = v_i \otimes \chi^{\#}(T)$ for every i. Thus $v_i \in V_\chi$, proving (1).

Conversely, assume that V_χ is G-stable for every representation of G and every character χ of H. By Chevalley's theorem, choose a finite dimensional representation (V, r) and a one dimensional subspace $L \subseteq V$ with $\mathrm{Stab}_G(L) = H$. The action of H on L is a one dimensional representation, hence a character

$$\chi : H \longrightarrow \mathrm{GL}(L) \simeq \mathbb{G}_m.$$

[10]It just means taking $R = \mathcal{O}(H) \otimes \mathcal{O}(G)$ and $g \in G(R)$ corresponding to $\mathcal{O}(G) \to \mathcal{O}(H) \otimes \mathcal{O}(G) : \alpha \mapsto 1 \otimes \alpha$ and $h \in H(R)$ corresponding to $\mathcal{O}(H) \to \mathcal{O}(H) \otimes \mathcal{O}(G) : \beta \mapsto \beta \otimes 1$.

Write $L = kv$. The restricted coaction is

$$\rho_{r|_H}(v) = v \otimes \chi^{\#}(T),$$

so $v \in V_\chi$.

By the assumed G-stability of V_χ, for all $R \in \mathrm{Alg}_k$, $g \in G(R)$ and $h \in H(R)$,

$$r_R(h)r_R(g)(v \otimes 1) = r_R(g)(v \otimes 1)\chi_R(h).$$

Applying $r_R(g^{-1})$ gives

$$r_R(g^{-1}hg)(v \otimes 1) = (v \otimes 1)\chi_R(h).$$

Because $\chi_R(h) \in R^\times$, this says that $g^{-1}hg$ stabilizes $L \otimes R$. As $\mathrm{Stab}_G(L) = H$, we have $g^{-1}hg \in H(R)$. Thus $H(R)$ is normal in $G(R)$ for every R, and H is normal in G. The connectedness assumption was used in the forward implication. □

Exercise A.12. Find a finite dimension representation V of GL_n and a 1 dimensional subspace W of V such that $\mathrm{Stab}_G(W) = \mathrm{SL}_n$.

Proof. Consider

$$V = ke \oplus \bigwedge^n k^n,$$

with GL_n acting trivially on the first summand and by det on the second. This gives a representation $r : \mathrm{GL}_n \to \mathrm{GL}(V)$. Let $\omega = e_1 \wedge \cdots \wedge e_n$ be the standard volume vector and take

$$W = k(e + \omega) \subseteq ke \oplus \bigwedge^n k^n.$$

For a k-algebra R, write $g = (g_{ij}) \in \mathrm{GL}_n(R)$. Then

$$r_R(g)(e + \omega) = e + \det(g)\omega.$$

This generates the same rank-one summand as $e + \omega$ exactly when $\det(g) = 1$ (compare the first coordinates). Therefore the stabilizer functor is represented by

$$\mathcal{O}(\mathrm{GL}_n)/(\det -1) = \mathcal{O}(\mathrm{SL}_n),$$

so $\mathrm{Stab}_G(W) = \mathrm{SL}_n$. □

A.4 Week 4-5

Exercise A.13. Show that all finite dimensional representations of $\mathbb{G}_m \times \mathbb{G}_m$ are direct sum of eigenspaces.

Solution. Let $r : \mathbb{G}_m \times \mathbb{G}_m \to \mathrm{GL}_V$ be a representation, where V is a finite dimensional k-vector space. We want to show that

$$V = \bigoplus_{\chi:\mathbb{G}_m\times\mathbb{G}_m\to\mathbb{G}_m} V_\chi.$$

Recall that

$$V_\chi = \{v \in V \mid \rho_r(v) = v \otimes \chi^\#(T)\}.$$

For every character χ, the induced map is

$$\chi^\# : k[T, T^{-1}] \longrightarrow k[x, x^{-1}, y, y^{-1}].$$

Since T must map to a unit, initially $\chi^\#(T) = cx^m y^n$ for some $c \in k^\times$ and $m, n \in \mathbb{Z}$. As $\chi(e) = e$, evaluating at the identity gives $c = 1$. Denote by $\chi_{m,n}$ the character with $\chi^\#_{m,n}(T) = x^m y^n$.

Write the coaction as

$$\rho_r : V \longrightarrow V \otimes \mathcal{O}(\mathbb{G}_m \times \mathbb{G}_m), \qquad \rho_r(v) = \sum_{m,n\in\mathbb{Z}} v_{m,n} \otimes x^m y^n,$$

where the sum is finite. We claim that $v_{m,n} \in V_{\chi_{m,n}}$. The comodule identity says that

$$(\rho_r \otimes \mathrm{id}) \circ \rho_r = (\mathrm{id} \otimes m^\#) \circ \rho_r.$$

Recall that $m^\#(f)(g_1, g_2) = f(g_1 g_2)$. In our situation,

$$m^\#(x^m y^n)((a, b), (c, d)) = (ac)^m (bd)^n,$$

so $m^\#(x^m y^n) = x^m y^n \otimes x^m y^n$. Therefore

$$\sum_{m,n} \rho_r(v_{m,n}) \otimes x^m y^n = \sum_{m,n} v_{m,n} \otimes x^m y^n \otimes x^m y^n.$$

The Laurent monomials $\{x^m y^n\}_{m,n\in\mathbb{Z}}$ are k-linearly independent. Hence

$$\rho_r(v_{m,n}) = v_{m,n} \otimes x^m y^n, \qquad v_{m,n} \in V_{\chi_{m,n}}.$$

Thus $V = \sum_{m,n} V_{\chi_{m,n}}$. A sum of character eigenspaces is direct, so

$$V = \bigoplus_{m,n\in\mathbb{Z}} V_{\chi_{m,n}}.$$

Only finitely many summands are nonzero because $\dim_k V < \infty$. □

Exercise A.14. Let $G = \mathrm{SL}_2$ and let B be the closed subgroup of G of the upper triangular matrices. Let U be the closed subgroup of the unipotent matrices of B.

1. Show that every character of $U \simeq \mathbb{G}_a$ is trivial.
2. Let $G = \mathrm{SL}_2$ and let B be the closed subgroup of G of the upper triangular matrices. Deduce the characters of B over k.
3. Consider the regular representation $\mathcal{O}(G)$ of G. Consider $\mathcal{O}(G)$ as a representation of U. Compute the eigenspace $\mathcal{O}(G)^U$ of U in $\mathcal{O}(G)$.
4. Determine the characters of B that occurs in the regular representation of G. Deduce the possible characters of B that occurs in some finite dimensional representation of G.

Solution. **(a)** Let $\chi : \mathbb{G}_a \to \mathbb{G}_m$ be a character. Its comorphism is

$$\chi^{\#} : k[T, T^{-1}] \longrightarrow k[x].$$

Since $\chi^{\#}(T) \in k[x]^\times = k^\times$ and

$$\chi^{\#}(T)(e_{\mathbb{G}_a}) = T(\chi(e_{\mathbb{G}_a})) = T(e_{\mathbb{G}_m}) = 1,$$

we have $\chi^{\#}(T) = 1$. Thus χ factors through $\mathbb{G}_a \to \operatorname{Spec} k \xrightarrow{e} \mathbb{G}_m$ and is trivial.

(b) We have

$$B = \operatorname{Spec}\big(k[x, y, z]/(xy - 1)\big), \qquad \begin{pmatrix} x & z \\ 0 & y \end{pmatrix}, \quad xy = 1.$$

A character $\chi : B \to \mathbb{G}_m$ corresponds to

$$\chi^{\#} : k[T, T^{-1}] \longrightarrow k[x, y, z]/(xy - 1).$$

The element $\chi^{\#}(T)$ must be a unit. Combining this with $\chi^{\#}(T)(e_B) = T(e_{\mathbb{G}_m}) = 1$, we

obtain

$$\chi^{\#}(T) = \bar{x}^{\,n} \quad (n \in \mathbb{Z}), \qquad \bar{x} = \bar{y}^{-1}.$$

Geometrically, every character has the form

$$\chi_n : \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \longmapsto a^n, \qquad n \in \mathbb{Z}.$$

(c) Let $\iota : U \hookrightarrow G$ be the inclusion and let $r : U \xrightarrow{\iota} G \to \mathrm{GL}_{\mathcal{O}(G)}$ be the restriction of the regular representation. Its coaction is

$$\rho_r = (\mathrm{id} \otimes \iota^{\#}) \circ m^{\#} : \mathcal{O}(G) \longrightarrow \mathcal{O}(G) \otimes \mathcal{O}(U).$$

Thus

$$\mathcal{O}(G)^U = \{v \in \mathcal{O}(G) \mid \rho_r(v) = v \otimes 1\}.$$

For $\alpha \in G(k)$, $\beta \in U(k)$, and $v \in \mathcal{O}(G)^U$,

$$v(\alpha) = (v \otimes 1)(\alpha, \beta) = \rho_r(v)(\alpha, \beta) = m^{\#}(v)(\alpha, \beta) = v(\alpha\beta).$$

Consequently, for every $\lambda \in k$,

$$v\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} = v\begin{pmatrix} t_{11} & t_{11}\lambda + t_{12} \\ t_{21} & t_{21}\lambda + t_{22} \end{pmatrix}.$$

We claim that

$$\mathcal{O}(G)^U = k[\bar{T}_{11}, \bar{T}_{21}] \subseteq \mathcal{O}(G) = k[T_{ij}]/(\det(T_{ij}) - 1),$$

the subring generated by $\bar{T}_{11}$ and $\bar{T}_{21}$.

First,

$$\mathcal{O}(G)_{\bar{T}_{11}} = k[\bar{T}_{11}, \bar{T}_{11}^{-1}, \bar{T}_{12}, \bar{T}_{21}].$$

View v as an element of this ring; on $D(\bar{T}_{11})$, this means

$$v(t_{11}, t_{12}, t_{21}) = v\left(t_{11}, t_{12}, t_{21}, \frac{t_{12}t_{21} + 1}{t_{11}}\right).$$

Then

$$v(t_{11}, t_{12}, t_{21}) = v(t_{11}, t_{12} + \lambda t_{11}, t_{21}) \quad \text{for every } \lambda \in k.$$

Hence $v \in k[\bar{T}_{11}, \bar{T}_{11}^{-1}, \bar{T}_{21}]$. Similarly, view v as an element of

$$\mathcal{O}(G)_{\bar{T}_{21}} = k[\bar{T}_{21}, \bar{T}_{21}^{-1}, \bar{T}_{11}, \bar{T}_{22}].$$

Here

$$v(t_{11}, t_{21}, t_{22}) = v\left(t_{11}, \frac{t_{11}t_{22} - 1}{t_{21}}, t_{21}, t_{22}\right),$$

and

$$v(t_{11}, t_{21}, t_{22}) = v(t_{11}, t_{21}, t_{21}\lambda + t_{22}).$$

Thus $v \in k[\bar{T}_{11}, \bar{T}_{21}, \bar{T}_{21}^{-1}]$. Therefore

$$\begin{aligned} v &\in k[\bar{T}_{11}, \bar{T}_{21}, \bar{T}_{21}^{-1}] \cap k[\bar{T}_{11}, \bar{T}_{21}, \bar{T}_{11}^{-1}] \cap \mathcal{O}(G) \\ &= k[\bar{T}_{11}, \bar{T}_{21}], \end{aligned}$$

which proves the claim.

(d) First, let $v \in \mathcal{O}(G)_{\chi_n}$. For $\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} \in G(k)$ and $\begin{pmatrix} \lambda & \mu \\ 0 & \lambda^{-1} \end{pmatrix} \in B(k)$, the eigenspace condition gives

$$\begin{aligned} \lambda^n v\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} &= (v \otimes \chi_n^{\#}(T))\left(\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix}, \begin{pmatrix} \lambda & \mu \\ 0 & \lambda^{-1} \end{pmatrix}\right) \\ &= \rho_r(v)\left(\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix}, \begin{pmatrix} \lambda & \mu \\ 0 & \lambda^{-1} \end{pmatrix}\right) \\ &= m^{\#}(v)\left(\begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix}, \begin{pmatrix} \lambda & \mu \\ 0 & \lambda^{-1} \end{pmatrix}\right) \\ &= v\begin{pmatrix} t_{11}\lambda & t_{11}\mu + t_{12}\lambda^{-1} \\ t_{21}\lambda & t_{21}\mu + t_{22}\lambda^{-1} \end{pmatrix}. \end{aligned}$$

The same argument as above gives $v \in k[\bar{T}_{11}, \bar{T}_{21}]$. Analyzing the dependence on λ, v must be homogeneous of degree n. Thus

$$\mathcal{O}(G)_{\chi_n} = \begin{cases} 0, & n < 0, \\ \bigoplus\limits_{\substack{p,q \geq 0 \\ p+q=n}} k\bar{T}_{11}^p \bar{T}_{21}^q, & n \geq 0. \end{cases}$$

Second, for $n \geq 0$, we can extend this eigenspace to a G-stable, finite dimensional k-subspace V of $\mathcal{O}(G)$. Then $B \to G \to \mathrm{GL}_V$ admits χ_n. Conversely, if $B \to G \to \mathrm{GL}_V$ admits χ_n for some finite dimensional representation (V, r), then (V, r) can be embedded into

$$\left(\mathcal{O}(G)^{\oplus \dim_k V}, (m^{\#})^{\oplus \dim_k V}\right).$$

By the preceding computation, we must have $n \geq 0$. Therefore χ_n occurs in some finite dimensional representation of G if and only if $n \geq 0$. □

Exercise A.15. Let p be a prime number.

1. Let G be an abstract finite group. For $x \in G$, show that there exists unique elements $x_{p'}, x_p \in G$ such that $x = x_{p'}x_p = x_p x_{p'}$ with the order of $x_{p'}$ coprime to p and the order of x_p a power of p.
2. Let $G = \mathrm{GL}_n(\mathbb{F}_p)$. For $x \in G$, show that $x = x_{p'}x_p$ is the Jordan decomposition of x with $x_{p'}$ the semisimple part and x_p the unipotent part.
3. Determine the Jordan decomposition for $\begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F}_5)$.

Solution. **(a)** Write $\mathrm{ord}(x) = p^m n$, where $\gcd(p, n) = 1$. Choose $a, b \in \mathbb{Z}$ such that $p^m a + nb = 1$.

Existence. We have

$$x = x^{p^m a}x^{nb} = x^{nb}x^{p^m a}.$$

Moreover,

$$\mathrm{ord}(x^{p^m a}) = \frac{p^m n}{\gcd(p^m a, p^m n)} = n, \qquad \mathrm{ord}(x^{nb}) = \frac{p^m n}{\gcd(nb, p^m n)} = p^m,$$

since $\gcd(a, n) = 1$ and $\gcd(p^m, b) = 1$. Set $x_p = x^{nb}$ and $x_{p'} = x^{p^m a}$.

Uniqueness. Suppose that

$$x = x_p x_{p'} = x_{p'}x_p = y_p y_{p'} = y_{p'}y_p$$

is another such decomposition. Then

$$xy_p = y_p y_{p'} y_p = y_p y_p y_{p'} = y_p x,$$

and similarly $xy_{p'} = y_{p'}x$. Since $x_p = x^{nb}$ and $x_{p'} = x^{p^m a}$, these elements also commute with y_p and $y_{p'}$. Thus $x_p, x_{p'}, y_p, y_{p'}$, and x all commute. Since $\gcd(\mathrm{ord}(y_p), \mathrm{ord}(y_{p'})) = 1$,

$$\mathrm{ord}(x) = \mathrm{ord}(y_p)\,\mathrm{ord}(y_{p'}),$$

so $\operatorname{ord}(y_p) = \operatorname{ord}(x_p)$ and $\operatorname{ord}(y_{p'}) = \operatorname{ord}(x_{p'})$. Consequently,

$$\begin{aligned}\operatorname{ord}(y_p^{-1}x_p) \mid \operatorname{lcm}(\operatorname{ord}(y_p), \operatorname{ord}(x_p)) = \operatorname{ord}(x_p),\\ \operatorname{ord}(y_{p'}x_{p'}^{-1}) \mid \operatorname{ord}(x_{p'}).\end{aligned}$$

But $y_p^{-1}x_p = y_{p'}x_{p'}^{-1}$ by the decompositions. Its order divides both a power of p and an integer coprime to p, so

$$y_p^{-1}x_p = y_{p'}x_{p'}^{-1} = e.$$

Hence $y_p = x_p$ and $y_{p'} = x_{p'}$.

(b) The element x_p is unipotent, since in characteristic p,

$$(x_p - \mathrm{id})^{p^m} = x_p^{p^m} - \mathrm{id} = \mathrm{id} - \mathrm{id} = 0.$$

The element $x_{p'}$ satisfies $x_{p'}^r - \mathrm{id} = 0$ for some r with $\gcd(r, p) = 1$. Let $P(T) \in \mathbb{F}_p[T]$ be its minimal polynomial. Then $P(T) \mid T^r - 1$. The polynomial $T^r - 1$ has no multiple roots in $\overline{\mathbb{F}}_p$ because $\gcd(r, p) = 1$. Thus $P(T)$ has no multiple roots in $\overline{\mathbb{F}}_p$, so $x_{p'}$ is diagonalizable over $\overline{\mathbb{F}}_p$, i.e., $x_{p'}$ is semisimple.

(c) Put $x = \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}$. All the following matrix equalities are in $\mathbb{F}_5$:

$$\begin{aligned}x^2 &= \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} 10 & 12 \\ 12 & 18 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 2 & 3 \end{pmatrix},\\ x^3 &= \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 0 & 2 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 6 & 11 \\ 6 & 15 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},\\ x^4 &= \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 6 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}.\end{aligned}$$

Then

$$x^5 = \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 7 & 10 \\ 15 & 12 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix},$$

and hence

$$x^{20} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}^4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Consider

$$x_p = x^{-4} = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & 4 \\ 4 & 4 \end{pmatrix}, \qquad x_{p'} = x^5 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

□

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